

Testing Strategy and Hands-on Testing Guide

How the product is tested, how to run every layer, how to sign in as each user type, and where to go to test each feature — with screenshots and the expected result.

Fixture-only. Every name, number, e-mail and password in this guide is an invented, published constant from the fixture seed. Nothing here is a secret and nothing here is real client data. Do not enter real data into a fixture-only environment.

Audience — testers, operators and reviewers of the QMULATE Nazarah platform (internal).

Provenance — Written 2026-09-09 at `main` after the Sprint 12 merge (`b87552b` ; CI run `34369010748` , all eight jobs green). Screenshots were taken on the fixture seed of that commit. Everything in this document is fixture-only: every name, number, e-mail and password below is a published, invented constant from `data/fixtures/sample-waqf.json` and `packages/database/src/seed/` . Nothing here is a secret, and nothing here is real client data.

Printed — 2026-09-09. Source of truth: `docs/testing/TESTING-GUIDE.md` in the repository; when the two disagree, the repository wins.

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This document does two jobs:

1. **Testing strategy** — what the automated suites are, what each layer proves, how to run them, how to read a CI verdict, and where the coverage gaps are (§1–§3, §9).
 2. **Hands-on tutorial** — how to stand up the app locally, sign in as each seeded user type, and where to go to test each feature, with screenshots and the expected result (§4–§8).
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1 · The testing pyramid as it exists in this repo

E2E (Playwright, ar + en)	14 spec files · 49 tests × 2 locales = 98
Integration (real PostgreSQL)	96 files: database + api + jobs + worker
Guardrail (G-8, subprocess-driven)	2 files: residency + importer
Unit (vitest, per workspace, no database)	96 files across 11 workspaces

Layer	Where	What it proves	Runs against
Unit	<code>packages/*/test</code> , <code>apps/web/test</code> , <code>packages/domain/src/**/__tests__</code>	Pure logic: the distribution engine (thousands of enumerated runs), the compliance/deadline engine, the access matrix, i18n catalogue parity, the Arabic floor, schema/vocabulary parity pins, env schema, rate-limit posture	nothing external
Integration	<code>*.integration.test.ts</code> in <code>packages/database</code> , <code>packages/api</code> , <code>packages/jobs</code> , <code>apps/worker</code>	The database as a <i>constraint system</i> : guard triggers, privilege separation (ADR-0008), write-once Shart, append-only audit, maker≠checker, the approval-decision plane, onboarding gates, birth admission, the importer; the tRPC procedure ladder and scoping	a migrated + seeded PostgreSQL with the three roles provisioned
Guardrail (G-8)	<code>packages/database/test/residency-guardrail.test.ts</code> , <code>import-guardrail.test.ts</code>	Seed refuses anything but <code>fixture-only</code> ; importer refuses anything but <code>production + ksa</code> ; never both	subprocesses of the real seed/import CLIs
E2E	<code>apps/web/e2e/*.spec.ts</code>	What a <i>screen shows</i> to a <i>seated</i> user, in both locales: the journeys in §7, deny-by-default, RTL, no raw keys leaking	a production <code>next start</code> build over a seeded database

Why the pyramid is shaped this way. The product's invariants (corpus never distributed, Shart immutable, `SHART_INCOMPLETE` halts, maker ≠ checker, access matrix holds) are enforced in the **database** with `ENABLE ALWAYS` triggers and privilege separation, not in application code. So the integration layer is unusually heavy and unusually important: a unit test of a TypeScript function cannot prove a trigger fires. E2E is deliberately thin — one journey per feature, both locales — because its job is to prove the *screen* tells the truth, not to re-prove the engine.

Two verification rules the suites follow, and manual testers should too:

- **Green positive control before every negative.** A refusal is only evidence if the same path *succeeds* for the entitled seat in the same run. Every suite asserts both halves; when you test by hand, do the same (§7 pairs each negative with its positive).
- **A claim about what a screen shows is a different class from a claim about what a function returns.** The API can be right and the page can still render a zero where it should say "read refused". Check the words on the screen.

2 · Running the suites

All commands run from the repo root. There is no Docker on the reference machine; `scripts/dev-postgres.ts` runs an embedded PostgreSQL. CI uses a real `postgres:16-alpine` service container and is the source of truth.

2.1 Fast lane (no database) — run before every push

```
pnpm typecheck && pnpm lint && pnpm format:check && pnpm test
```

`pnpm test` is turbo-driven across **all** workspaces. Do not hand-pick packages: S12's first CI red was a unit suite in `@qmulate/auth` that had not been run locally because six packages were run by hand and the seventh was skipped.

2.2 Integration — the canonical sequence

```
pnpm exec tsx scripts/dev-postgres.ts --name it --port 54461 --reset --run "pnpm exec tsx scripts/provision-db-roles.ts && pnpm --filter @qmulate/database run migrate:deploy && pnpm run db:seed && pnpm test:integration"
```

Rules that cost real time when broken:

- **--reset alone runs the suites against an unmigrated cluster** and reports dozens of files red on `42P01 relation does not exist`. That is the harness, not the code. The four-step chain above (provision → migrate → seed → test) is the only correct order.
- **Integration BEFORE E2E on the same cluster, or use --reset**. The E2E leg leaves state behind (born endowments, enrolled TOTP, cleared gates). Some tables are write-once (`42501`), so a second run on a dirty cluster fails for reasons that have nothing to do with the code.
- **Use isolated, named clusters** (`--name` , `--port`). Never share a cluster between two concurrent sessions.

2.3 Guardrail (G-8)

```
pnpm test:guardrail
```

No database needed; it spawns the seed and importer CLIs and asserts their refusals by exit code and message.

2.4 E2E — the one-command local leg

```
pnpm test:e2e:local-fixture-only
```

This resets a cluster, provisions, migrates, seeds, **builds** `apps/web` **cold**, starts `next start` on **port 3000**, and runs Playwright with `CI=1` in both locales. It refuses to start if port 3000 is taken. Port 3000 is the harness's port: **keep human-facing servers off it** (§4 uses 3100) and never set `reuseExistingServer`.

The E2E server is the *production* build on purpose: `next dev` restarts itself when the V8 heap crosses 80 %, and for seven CI runs that restart was silently absorbed by a retry. `apps/web/e2e/web-server.mjs` now fails the run if that ever happens again.

2.5 The CI verdict

CI has eight jobs: Typecheck · Lint · Unit tests · Build · Integration tests · Residency guardrail (G-8) · E2E (Playwright, ar + en) · Deploy → staging (Railway). Take the verdict from the API, never from the log and never from a job's conclusion alone:

```
gh run view <run-id> --json status,conclusion,jobs --jq '.status, .conclusion, (.jobs[] | "\n(.name): \(.status)/\(.conclusion)")'
```

The deploy job reports success on main while deploying nothing. Its guard step passes and then *Enable corepack / setup-node / Install dependencies / Deploy / Apply migrations on staging* are all skipped because no Railway token is configured. Read it **per step**:

```
gh run view <run-id> --json jobs --jq '.jobs[] | select(.name|startswith("Deploy")) | .steps[] | "\(.name) → \(.conclusion)''
```

A push to a branch **cancels the in-flight run** for that branch (`cancel-in-progress: true`). If you need a run's evidence, wait for it before pushing again.

3 · What each gate and scenario is proven by

The Definition of Done (docs/product/prd/17-build-ship-dod.md) names release gates G-1...G-10 and verification scenarios V-1...V-12. Where they live:

Gate / scenario	Proven by	Manual check (§7)
G-1 Audit immutable	database integration: UPDATE / DELETE on audit_event refused at the DB	— (no UI; the trail is read via the API)
G-2 No commingling	migration 19's composite FKs + integration suite	Financials screen: every receipt on a dedicated account (§7.8)
G-3 Maker ≠ checker	database + api integration (SEGREGATION_OF_DUTIES), distribution.spec.ts	Submit a run as the accountant, approve as a <i>different</i> Nazir (§7.7)
G-4 KYC gates distribution	domain engine tests, distribution.spec.ts (KYC_UNVERIFIED line withheld)	Wizard lines step on waqf-001 , ben-003 is Withheld (§7.7)
G-5 Statutory deadlines	domain calendar tests, compliance-journey.spec.ts	Dashboard board for waqf-003 is red on the overdue certificate update (§7.1)
G-6 AML no-tipping-off	database + api integration (compartment subtraction)	GOV-AML-02 is invisible on every register outside the compartment
G-7 Access matrix holds	kernel.spec.ts , endowment.spec.ts , grant-escalation and router-introspection pins	Sign in as unscoped@ and beneficiary.ben-001@ (§7.9)
G-8 Residency guardrail	pnpm test:guardrail + CI steps G-8a...f	Run the importer CLI locally and watch it refuse (§8)
G-9 Distribution integrity	~2,160 domain tests incl. enumerated runs; invariants I1-I8, I-R1	Wizard review step lists the invariants asserted (§7.7)
V-1 corpus excluded from a run	distribution.spec.ts	rev-005 (istibdal proceeds) visibly excluded on the review step
V-11 Gate 02 blocks a run and a filing	onboarding-gate.spec.ts , api + database gate suites	§7.5
V-12 importer refuses a non-KSA / non-production target	import-guardrail.test.ts (15) + import-apply integration (4)	§8

4 · Stand up a local environment for manual testing

Prerequisites: Node 22, pnpm (corepack), `pnpm install` done once. No Docker needed.

One command starts a throwaway PostgreSQL, provisions the three database roles, applies all 54 migrations, seeds the fixture, and starts the web app on **port 3100**:

```
pnpm exec tsx scripts/dev-postgres.ts --name manual --port 54460 --reset --run "pnpm exec tsx
scripts/provision-db-roles.ts && pnpm --filter @qmulate/database run migrate:deploy && pnpm
run db:seed && env -u MIGRATOR_DATABASE_URL -u SUPERUSER_DATABASE_URL -u PGBOSS_DATABASE_URL
BETTER_AUTH_URL=http://localhost:3100 NEXT_PUBLIC_APP_URL=http://localhost:3100 PORT=3100
pnpm --filter web exec next dev --port 3100"
```

Then open <http://localhost:3100/en/sign-in> (or `/ar/sign-in`).

Why the command looks like that:

- `dev-postgres.ts --run` injects five database URLs into the child (`DATABASE_URL` for `qmulate_app` , `ACCESS_MATRIX_DATABASE_URL` , `MIGRATOR_DATABASE_URL` , `SUPERUSER_DATABASE_URL` , `PGBOSS_DATABASE_URL`) and pins `DATA_CLASSIFICATION=fixture-only` .
- The migrate and seed steps need the owner/superuser URLs. **The web app refuses to boot if it can see them** (ADR-0008 round 6, `apps/web/src/instrumentation.ts`) — that is a designed guard, not a bug — so `env -u ...` strips the three non-runtime credentials before `next dev` starts. If you forget, the log says `apps/web refuses to boot: privilege separation` and the command exits 0.
- Port 3100, not 3000: 3000 belongs to the Playwright harness (§2.4).
- Re-running **without** `--reset` keeps your data (enrolled TOTP, cleared gates, born endowments). Re-running **with** `--reset` gives you the pristine fixture again.

The seed prints what it wrote. On this commit: 14 users, 12 credential accounts, 1 client, 3 founders, 6 endowments, 30 beneficiaries, 9 transactions, 1 historical distribution, 50 compliance obligations, 4 deadlines, 47 access grants, 1 pending approval request; 283 audited writes. It also prints the admin login and the one sentence you need for §5: *"TOTP enrolment is required on first sign-in."*

5 · Sign in: password + TOTP (every account, every time)

Every seeded operator shares one fixture password:

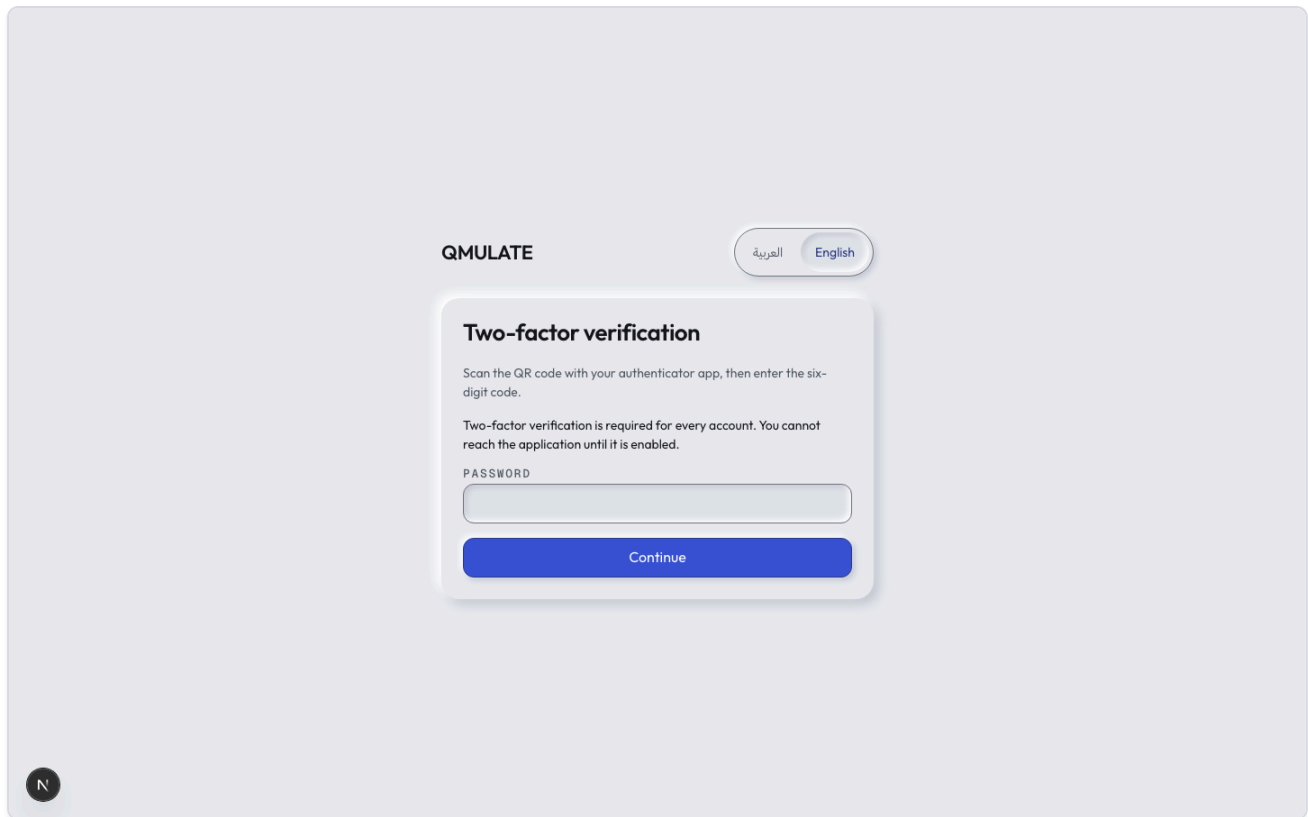
fixture-only-not-a-secret-9271

5.1 First sign-in enrolls a second factor (one-way door)

1. Open `/en/sign-in`, enter the e-mail and the password above, submit.

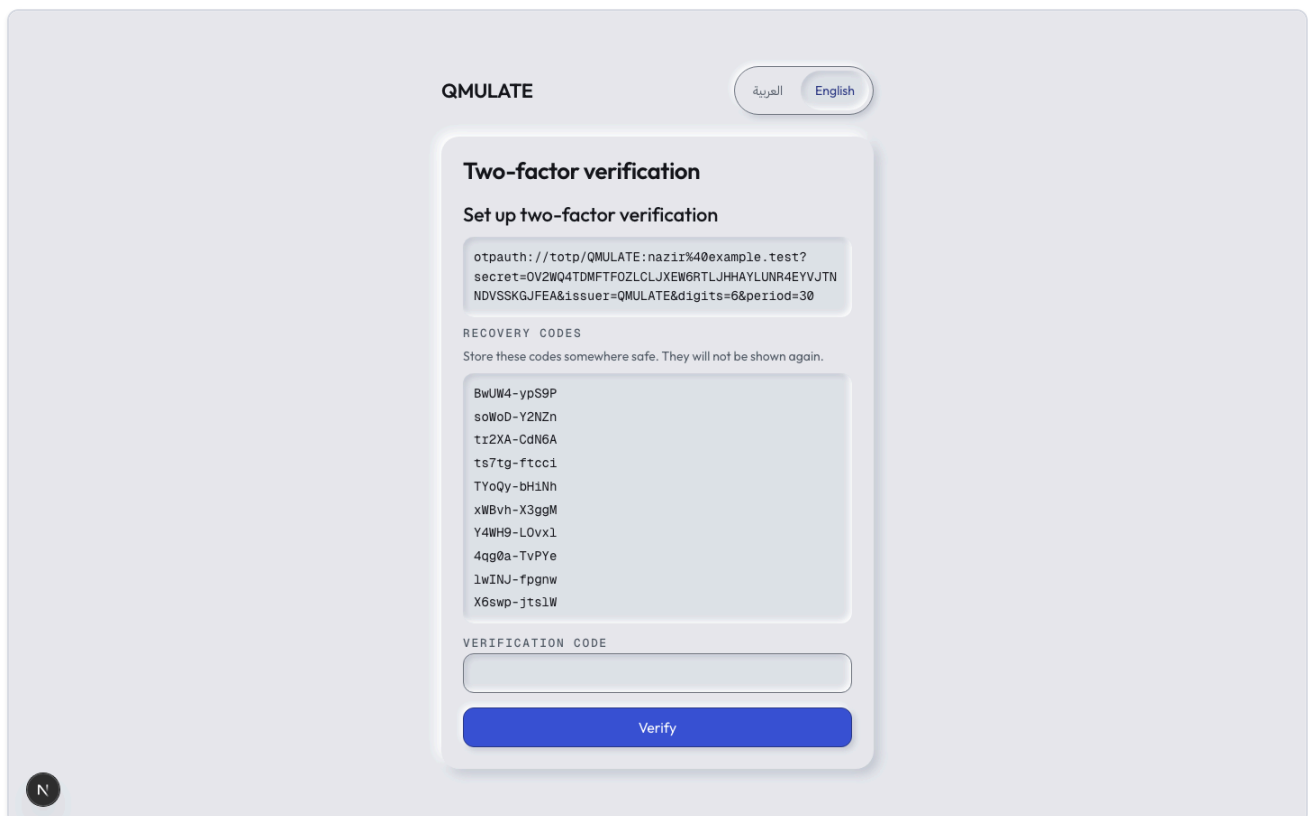
Sign-in

2. You land on `/en/two-factor` in the **enrol** phase. Two-factor is mandatory; you cannot reach the app without it. Re-enter the same password to start enrolment.



Two-factor — enrol, password

3. The page shows an `otpauth://` URI and ten recovery codes. Copy the recovery codes somewhere: they are shown once.



Two-factor — enrol, verify

4. Turn the URI into a six-digit code and submit it. Either:

- **Authenticator app:** in Google Authenticator / 1Password / Authy choose "enter a setup key" and paste the `secret=` value from the URI (the account name is the e-mail, issuer `QMULATE`, 6 digits, 30 s). The UI's copy says "scan the QR code", but the screen renders the URI as text, not as an image — pasting the secret is the working path today (see §9 gaps).
- **Helper script (this folder):**

```
node docs/testing/totp.mjs 'otpauth://totp/QMULATE:nazir%40example.test?secret=...
&issuer=QMULATE&digits=6&period=30'
```

prints the current code. Keep the URI; you need it on every later sign-in.

5. On success you are on `/en/dashboard`.

Enrolment cannot be repeated for the same user — better-auth returns the secret exactly once. If you lose the secret and the recovery codes, the only way back in as that user is `--reset` on the cluster (§4). Every later sign-in goes password → six-digit code.

5.2 Sign-up

`/en/sign-up` creates a new account. It has **no grants**, so it behaves exactly like `unscoped@example.test` (§7.9): it can sign in and enrol TOTP, and then is told nothing about the portfolio. Seats are issued by the matrix administrator through the API, not by self-service.

6 · The seeded users — who to sign in as

All twelve credential accounts use the password in §5. "Scope" is the set of endowments the seat holds a grant on; a screen for any other endowment answers *That record was not found* (`NO_GRANT` — deliberately indistinguishable from a non-existent record).

Sign in as	Role preset	Scope	Use it to test
<code>nazir@example.test</code>	NAZIR (42 verbs)	all six endowments	Everything an operator does: read every record, approve and sign runs, approve reserved matters, reopen a cleared onboarding gate, record maintenance policy. The primary trustee.
<code>approver@example.test</code>	NAZIR (second holder)	all six	The distinct checker . Approving a run the accountant submitted, when <code>nazir@</code> raised nothing. Two Nazir seats exist by product decision (PO-2): approval authority is role-level, maker ≠ checker is the invariant.
<code>accountant@example.test</code>	FINANCE	all six	The maker : compute, review and submit a distribution run; record receipts and bank movements (API); read financials. Cannot approve.
<code>matrix-admin@example.test</code>	SYSTEM_ADMIN	<code>waqf-001</code>	Register an endowment (intake) for the family it already serves; issue seats (API). The only holder of <code>admin:access_matrix:write</code> . Cannot approve or sign anything.
<code>clerk@example.test</code>	COMPLIANCE_OFFICER	<code>waqf-004</code>	Clear onboarding gates, record reserved-matter chain steps (principal consent, counsel review). The staff seat of the S12 journeys.
<code>compliance@example.test</code>	CASE_MANAGER preset, compliance duty	<code>waqf-003</code> , <code>waqf-004</code> , <code>waqf-007</code>	The compliance board in three states (red / open / not computable); discharging a registration duty on the endowment record.
<code>case-manager@example.test</code>	CASE_MANAGER	all six	Deed read ; endowment record work; distribution <i>initiate</i> without approve.
<code>auditor@example.test</code>	AUDITOR	<code>001</code> , <code>003</code> , <code>004</code> , <code>005</code> , <code>007</code>	Read-only financials ; every write refused; every refused read stated in words.
<code>board@example.test</code>	FAMILY_BOARD	all six + the client membership	The family's read-only view of its endowments and the approvals queue.
<code>beneficiary.ben-001@example.test</code>	BENEFICIARY	<code>waqf-001</code> , own record <code>ben-001</code> only	Scope walls: sees only its own endowment's deed and its own line items; never co-beneficiaries, never the ledger.

Sign in as	Role preset	Scope	Use it to test
admin@example.test	SYSTEM_ADMIN, no endowment grant	none	Platform-level admin. Proves that admin authority is not an endowment grant: endowment screens answer not-found. The E0 exit subject.
unscoped@example.test	none	none	The negative control . A valid session with zero grants must see nothing. Never give it a grant.

Two more seeded users have **no credential account by construction** and cannot sign in:

service.deadline-sweeper@example.test (the declared non-human service seat that runs the deadline sweep) and control.sweep-coverage@example.test (a sweep-coverage control). They exist so the worker's context can be built in-process, and so a test can prove nobody can log in as them.

6.1 The seeded endowments (the subjects you will point at)

One fictional client, *Al-Rashidi Family (fictional)*, three founders, six endowments. Certificate numbers are FAKE-100000n .

id	Certificate	Type / size	Why it exists
waqf-001	FAKE-1000001	family (dhurri), medium, ORDERED entitlement	The V-1 / G-9 subject. Has the historical run dist-001 , the corpus receipt rev-005 (istibdal proceeds, never distributable), a KYC-unverified beneficiary (ben-003), and the overdue REGISTER_30BD deadline.
waqf-002	FAKE-1000002	family, small, SHARED	The healthy registration (discharged on time); a deed silent on the Nazir fee.
waqf-003	FAKE-1000003	public charitable , large	Three charitable jahas at 40/30/30; the red compliance board (overdue certificate update, UPDATE_15BD).
waqf-004	FAKE-1000004	family, small, direct use	The S12 subject: Gate 02 open (V-11), the clerk's endowment, the reserved-matter chain journey, the direct-use financial statement.
waqf-005	FAKE-1000005	family, small, LINEAGE	Intake state : its مآل (reversion) clause is unread, so a distribution run halts (REVERSION_CLAUSE_UNREAD). Carries the 16-member, three-generation lineage tree.
waqf-007	FAKE-1000007	family, small, LINEAGE	The computing lineage sibling (four-member tree, zuhur_only). Registration anchor recorded but outside calendar coverage.

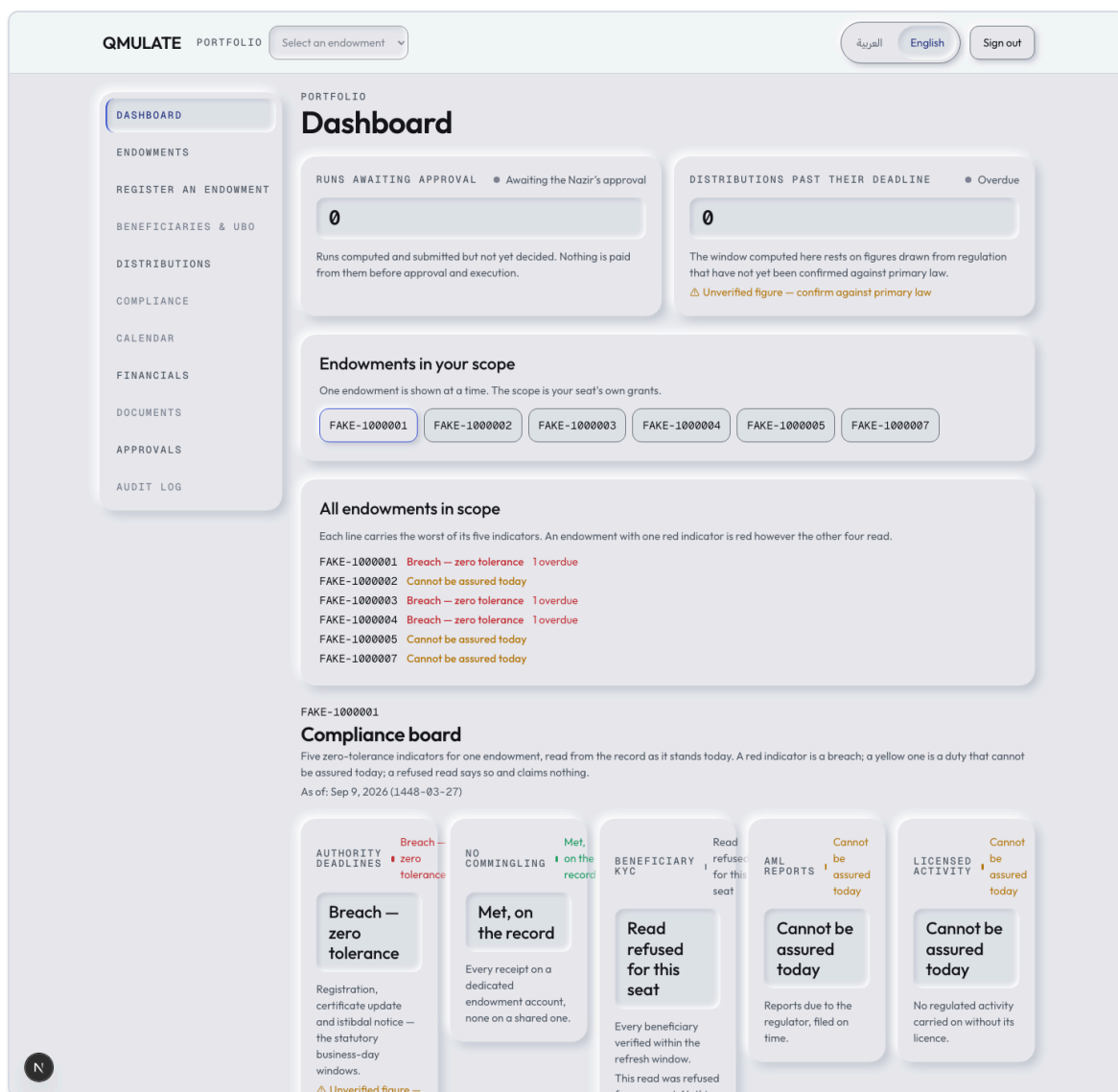
7 · Where to test what — feature by feature

Every route is locale-prefixed: swap `/en/` for `/ar/` to test the Arabic, right-to-left rendering of the same screen. The sidebar's greyed items (Beneficiaries & UBO, Compliance, Calendar, Documents, Audit Log) are rendered but **not linked** — they are not built (§9).

7.1 Dashboard and the compliance board

Route: `/en/dashboard` , `/en/dashboard?waqf=<id>` · **Who:** any seated user; what renders depends on the seat's verbs.

Sign in as `nazir@example.test` . The dashboard shows two KPI tiles (runs awaiting approval, distributions past their deadline — the latter flagged *unverified figure*), the endowments in scope, a one-line roll-up per endowment (each carries the worst of its five indicators), and the **compliance board** for the selected endowment: five zero-tolerance indicators, the statutory deadline table (one line per rule; a rule with no row says *why*), decisions required, KYC freshness, AML reporting, government filings.



Nazir — dashboard

What to check:

- Select `FAKE-1000003` (`waqf-003`). The Authority-deadlines indicator is **red**: *Certificate update (15 business days) — Overdue, due Jun 9, 2026 (1447-12-23)*. Every regulatory figure carries the `△ Unverified figure — confirm against primary law` marker (Binding rule 3).
- An indicator a seat cannot read says **"Read refused for this seat"** in words. It never renders a zero or a green light for a fact it was not allowed to read. Compare the same board as `compliance@example.test`:

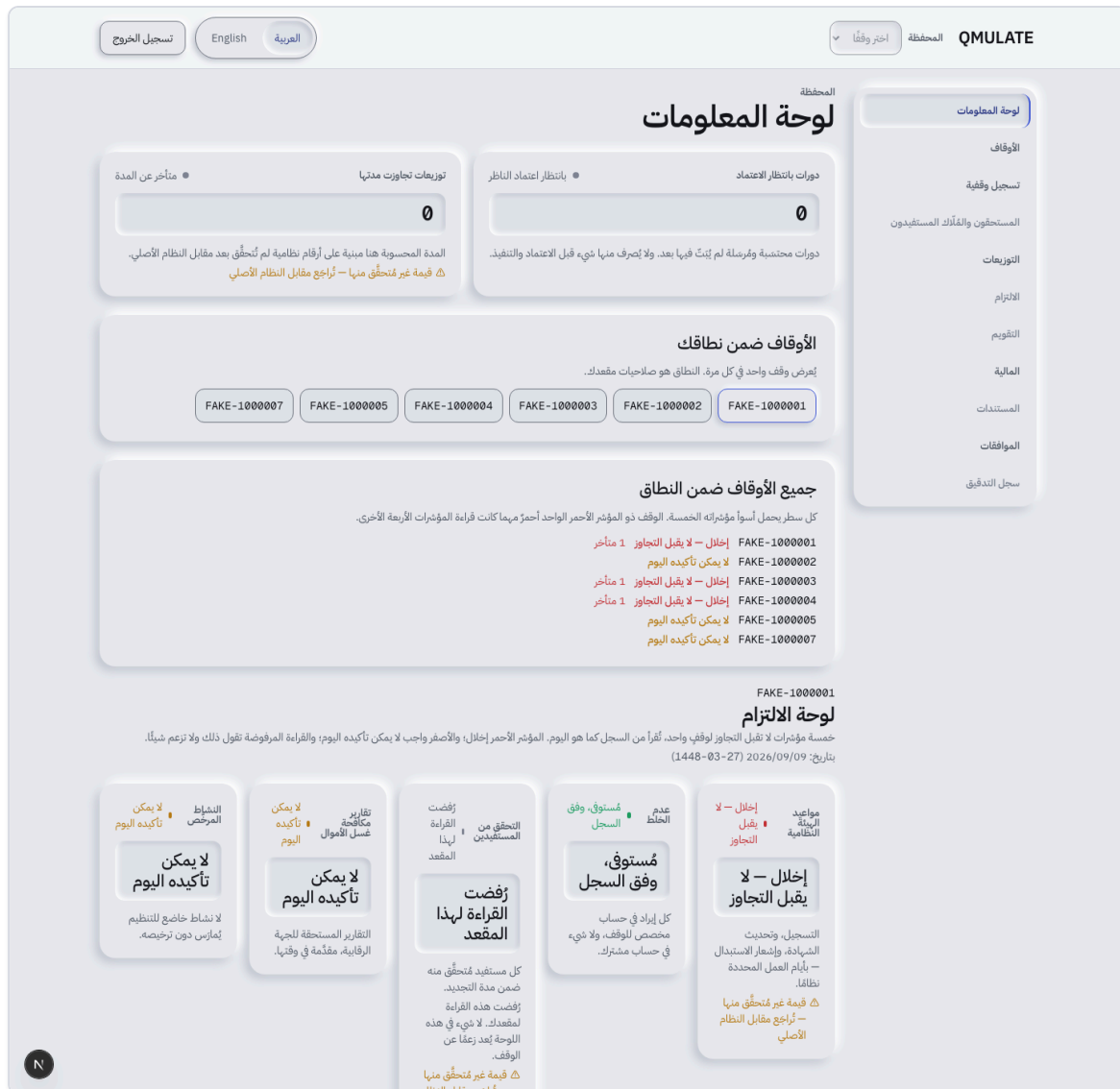
The screenshot shows the QMULATE PORTFOLIO Dashboard for the endowment `FAKE-1000003`. The interface is in English and includes a sidebar with navigation links: DASHBOARD, ENDOWMENTS, REGISTER AN ENDOWMENT, BENEFICIARIES & UBO, DISTRIBUTIONS, COMPLIANCE, CALENDAR, FINANCIALS, DOCUMENTS, APPROVALS, and AUDIT LOG. The main content area is titled "Dashboard" and includes several sections:

- RUNS AWAITING APPROVAL**: Shows 0 runs. Text: "Runs computed and submitted but not yet decided. Nothing is paid from them before approval and execution."
- DISTRIBUTIONS PAST THEIR DEADLINE**: Shows 0 distributions. Text: "The window computed here rests on figures drawn from regulation that have not yet been confirmed against primary law. Not recorded: 0. △ Unverified figure — confirm against primary law"
- Endowments in your scope**: Shows one endowment is shown at a time. The scope is your seat's own grants. Buttons for `FAKE-1000003`, `FAKE-1000004`, and `FAKE-1000007` are visible.
- All endowments in scope**: Lists three endowments with their status: `FAKE-1000003` (Breach — zero tolerance, 1 overdue), `FAKE-1000004` (Breach — zero tolerance, 1 overdue), and `FAKE-1000007` (Cannot be assured today).
- Compliance board**: A detailed view for `FAKE-1000003` showing five zero-tolerance indicators:
 - Authority Deadlines**: Breach — zero tolerance. Registration, certificate update and istbdal notice — the statutory business-day windows. △ Unverified figure — confirm against primary law
 - No Commingling**: Read refused for this seat. Every receipt on a dedicated endowment account, none on a shared one. This read was refused for your seat. Nothing in this panel is a claim
 - Beneficiary KYC**: Met, on the record. Every beneficiary verified within the refresh window. △ Unverified figure — confirm against primary law
 - AML Reports**: Cannot be assured today. Reports due to the regulator, filed on time.
 - Licensed Activity**: Cannot be assured today. No regulated activity carried on without its licence.

Compliance seat — red board on `waqf-003`

- `waqf-007` says *Cannot be assured today*: its registration anchor is recorded but falls outside the seeded holiday calendar, so the deadline is **not computable** — the board says so rather than guessing.
- Dual calendar everywhere: every date is Gregorian with the Hijri in parentheses.

Arabic: `/ar/dashboard` — the whole layout mirrors, the light source flips, Arabic labels are never uppercased or letter-spaced, and credential fields stay LTR islands.



Nazir — dashboard, Arabic

Automated: `compliance-journey.spec.ts` , `shell.spec.ts` , the compliance engine's unit suite.

7.2 Endowments: the record and its tabs

Route: `/en/endowments` → `/en/endowments/<waqfId>` · **Who:** any seat holding `endowment:waqf:read` on that endowment.

The list renders the hierarchy client → founder → endowment. The record page has tabs: **Record** · **Trusteeship deed** · **Classification** · **Founder's conditions** · **Beneficiaries** · **Reserved matters** · **Onboarding**. A tab the seat holds no verb for is refused on that tab while the others still read.

QMULATE

PORTFOLIO

Select an endowment

العربيةEnglishSign out

DASHBOARD

ENDOWMENTS

REGISTER AN ENDOWMENT

BENEFICIARIES & UBO

DISTRIBUTIONS

COMPLIANCE

CALENDAR

FINANCIALS

DOCUMENTS

APPROVALS

AUDIT LOG

PORTFOLIO

Endowments

Every endowment under trusteeship, grouped by the family that holds the engagement and the founder who established it.

CLIENT FAMILY

Al-Rashidi Family (fictional)

FOUNDER

Ibrahim Al-Rashidi (fictional)

FAKE-1000001

Family (dhurri) · In-kind (ayni)

Order of entitlement: Tiered — al-a'la fa-l-l-a'la

Open record

Medium

FAKE-1000007

Family (dhurri) · In-kind (ayni)

Order of entitlement: Lineage continuation

Open record

Small

FOUNDER

Sulaiman Al-Rashidi (fictional)

FAKE-1000002

Family (dhurri) · In-kind (ayni)

Order of entitlement: Shared — tashrik

Open record

Small

FAKE-1000005

Family (dhurri) · In-kind (ayni)

Order of entitlement: Lineage continuation

Open record

Small

FOUNDER

Aisha Al-Rashidi (fictional)

FAKE-1000003

Public charitable (khayri) · In-kind (ayni)

Order of entitlement: Shared — tashrik

Open record

Large

FAKE-1000004

Family (dhurri) · In-kind (ayni)

Order of entitlement: Not applicable — direct use of the asset

Open record

Small

Nazir — endowments

QMULATE

PORTFOLIO

Select an endowment

العربية

English

Sign out

DASHBOARD

ENDOWMENTS

REGISTER AN ENDOWMENT

BENEFICIARIES & UBO

DISTRIBUTIONS

COMPLIANCE

CALENDAR

FINANCIALS

DOCUMENTS

APPROVALS

AUDIT LOG

ENDOWMENT

FAKE-1000001

Medium

Founder (waqf): Ibrahim Al-Rashidi (fictional)

RECORD

TRUSTEESHIP DEED

CLASSIFICATION

FOUNDER'S CONDITIONS

BENEFICIARIES

RESERVED MAT

Identity and registration

CLIENT FAMILY	Al-Rashidi Family (fictional)
FOUNDER (WAQIF)	Ibrahim Al-Rashidi (fictional)
WAQF CERTIFICATE NUMBER	FAKE-1000001
WAQF DEED NUMBER	FAKE-DEED-455
AUTHORITY CLASSIFICATION	Medium
ENDOWMENT TYPE	Family (dhurri)
ENDOWMENT NATURE	In-kind (ayni)
REGISTRATION DATE	Mar 11, 1980 (1400-04-23)
CERTIFICATE VALID UNTIL	Not recorded
FISCAL YEAR END	12-31

Conditions carried on the deed

These are conditions of the founder's deed, not settings. Each is recorded once and then read; a correction is a superseding instrument.

ORDER OF ENTITLEMENT	Tiered — al-a'la fa-l-a'la
LINE-CONTINUATION STIPULATION	Sons' line only (zuhur)
CONDITIONS VERSION	1

Ultimate taker (maal al-waqf)

The deed records no ultimate taker

This is a positive statement read off the deed, not an omission.

Trusteeship

PRIMARY NAZIR	QMULATE (professional Nazir)
AUTHORIZED REPRESENTATIVE	Yes
JOINTLY AND SEVERALLY LIABLE	Yes

Nazir — waqf-001 record

Trusteeship deed (/deed): the primary Nazir and the authorised representative, joint-and-several liability stated. There is **no edit**: a recorded appointment is superseded by a court instrument, never edited (owner ruling, memo Q10). `case-manager@` and `beneficiary.ben-001@` can read this tab on `waqf-001` .

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QMULATE

PORTFOLIO

Select an endowment

العربية

English

Sign out

DASHBOARD

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REGISTER AN ENDOWMENT

BENEFICIARIES & UBO

DISTRIBUTIONS

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ENDOWMENT

FAKE-1000001

Medium

Founder (waqif): Ibrahim Al-Rashidi (fictional)

RECORD

TRUSTEESHIP DEED

CLASSIFICATION

FOUNDER'S CONDITIONS

BENEFICIARIES

RESERVED MATT

Primary Nazir

PRIMARY NAZIR

QMULATE (professional Nazir)

APPOINTED ON

Feb 1, 2025 (1446-08-02)

SUCCESSOR NAZIR

Not recorded

Authorized representative

NAME

Delegated Manager (fictional)

DELEGATED SCOPE

day-to-day property & finance ops

APPOINTED ON

Feb 15, 2025 (1446-08-16)

Joint and several liability

Yes

A delegated manager is jointly and severally liable with the Nazir.

A representative may initiate and act within the delegated scope. Approval and signature are not delegable: every act by the representative is recorded against the Nazir on whose behalf it was taken.

Eligibility

The conditions a trustee must meet before an appointment may be recorded. Each is captured and verified, never assumed.

Not yet verified

The conditions have been captured but no verification has been recorded. Captured flags record a claim, not a verification.

Primary Nazir

Meets the recorded conditions

Conditions

Muslim

Binding

Met

Full legal capacity

Binding

Met

Nazir — deed

Classification (/classification): size class and direct-use flag as two independent axes; the duties this class gates and the ones it excludes; **no settled band figure** (the SAR 200M / 50M thresholds are unverified and never printed as settled).

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FAKE-1000001

Medium

Founder (waqif): Ibrahim Al-Rashidi (fictional)

RECORD

TRUSTEESHIP DEED

CLASSIFICATION

FOUNDER'S CONDITIONS

BENEFICIARIES

RESERVED MATTER

Recorded classification

Medium

The bands this classification comes from

The value bands are configuration, not code: they are read from settings so a correction is a settings change.

Settings consulted

classification.threshold.large.sar classification.threshold.medium.sar

⚠ Unverified figure — confirm against primary law

Re-classification history

The history is append-only at the database. A recorded transition cannot be edited or removed.

No re-classification has been recorded for this endowment.

Duties that apply at this classification

Derived from the regulation's own catalogue and this endowment's recorded classification. The task instances and their statuses belong to the compliance module.

CODE	DUTY	WORKSTREAM	APPLIES TO
FIN-ACC-01	Subscribe to & maintain an approved accounting system	Financial	Every classification
FIN-ACC-03	Verify invoices/expenses before approval	Financial	Every classification
FIN-DIST-01	Compute & disburse ghalah per Shart via dedicated accounts DISTRIBUTE_3M_FYE	Financial	Excludes direct-utilization
FIN-DIST-02	Produce per-beneficiary distribution statements	Financial	Excludes direct-utilization
FIN-DIST-03	Observe waqif intent where the beneficiary is unspecified / naming lapses / shart is general; refer to the competent authority	Financial	Every classification
FIN-MGT-01	Maintain dedicated waqf account(s); no commingling	Financial	Every classification
ETH-MGT-	Prepare budget / estimate	Financial	Large and medium

Nazir — classification

Founder's conditions / Shart al-Waqif (/shart): the structured record, marked **immutable**. There is no edit control on this screen and no mutation at any API path. A "would halt" advisory shows what the engine would refuse on this deed.

QMULATE

PORTFOLIO

Select an endowment

العربية

English

Sign out

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ENDOWMENT

FAKE-1000001

Medium

Founder (waqf): Ibrahim Al-Rashidi (fictional)

RECORD

TRUSTEESHIP DEED

CLASSIFICATION

FOUNDER'S CONDITIONS

BENEFICIARIES

RESERVED MAT

This record cannot be edited

The founder's conditions are written once. No approval opens them — there is no such approval — and no correction, backfill or migration may alter them. That is the purpose of the record: to be what the founder instructed.

If a correction is lawful, it is a new record

Where a competent authority or a court directs a change, the answer is a superseding instrument recorded as a new record alongside this one. The original stays exactly as it stands.

Conditions as recorded

VERSION	1
RECORDED ON	Mar 11, 1980 (1400-04-23)
ORDER OF ENTITLEMENT	Tiered — al-a'la fa-l-a'la
LINE-CONTINUATION STIPULATION	Sons' line only (zuhur)
TIER NAMED BY THE DEED	1 الطبقة الأولى BUTUN ZUHUR 2 الطبقة الثانية ZUHUR
LINES NAMED BY THE DEED	BUTUN ZUHUR
MAINTENANCE RESERVE (SIYANA)	fixed
	Silent is not the same as none. A deed that stipulates no maintenance reserve and a deed nobody could read are different facts and are kept apart.
DISBURSEMENT CHANNEL	FAMILY
NAZIR FEE SET BY THE DEED	PERCENT_OF_REVENUE 10%
	Unverified figure — confirm against primary law

Completeness

What is missing here decides whether a distribution can be computed at all.

Nothing halting is missing from these conditions.

Advisory — a distribution would still run

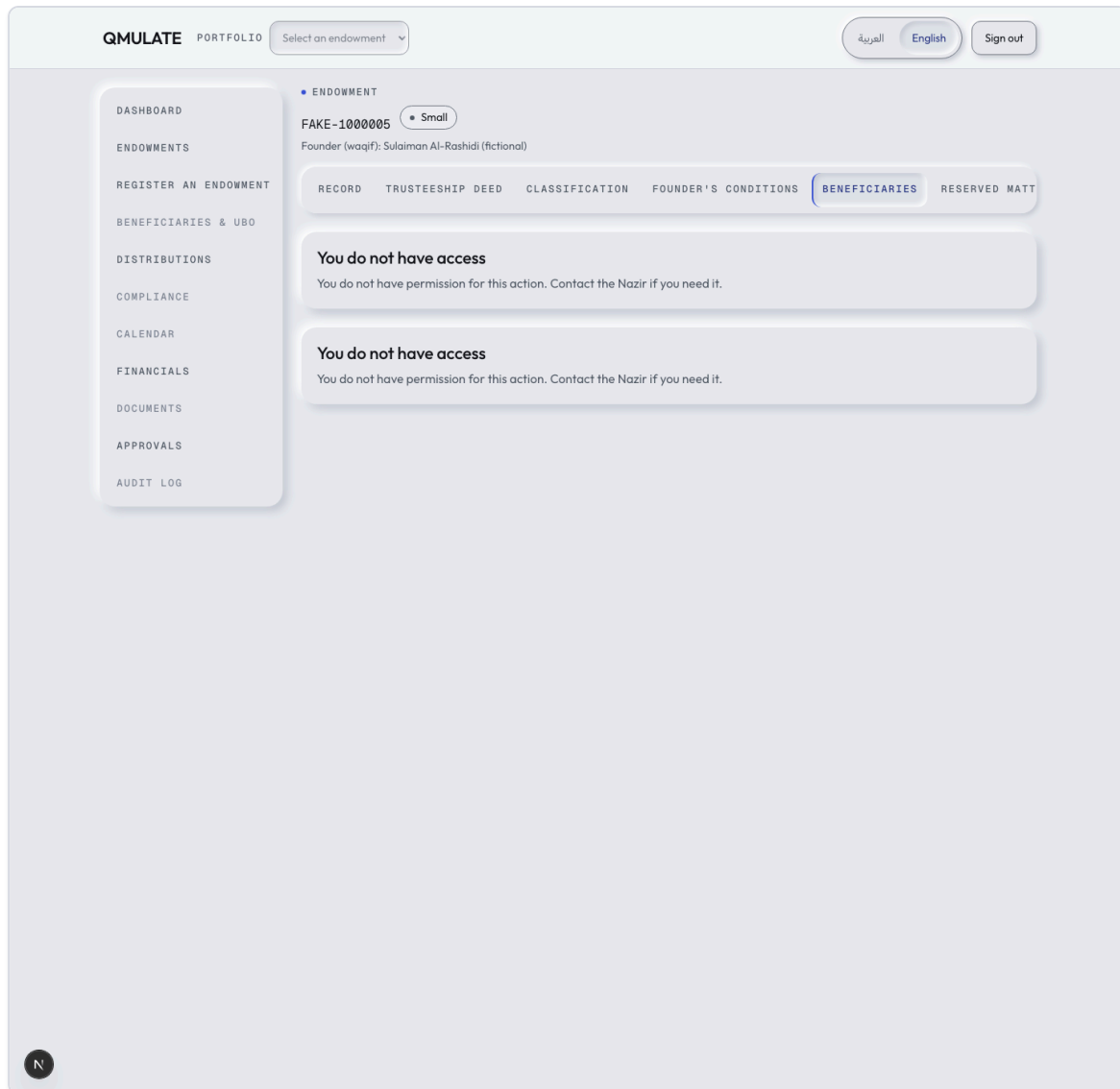
Worth cashing, but none of these halts a computation

Nazir — Shart al-Waqf

Beneficiaries (/beneficiaries): the registry, KYC labels, and the **lineage tree** derived from recorded parent edges. Use `waqf-005` for the 16-member, three-generation tree; the integrity panel is all-clear when derived tiers agree with the recorded ones. A `CATEGORY_ONLY` row without a captured category says *disbursement blocked* on the row.

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Nazir — waqf-005 beneficiaries and lineage

Negative controls to run alongside: as `board@example.test` the same record reads, but no create / reclassify / deed-term affordance exists on any endowment screen (there is none for the Nazir either — those acts are reserved matters). As `unscoped@` the list is empty and every record is not-found.

Automated: `endowment.spec.ts` , `endowment-journey.spec.ts` , `beneficiaries.spec.ts` , `kernel.spec.ts` .

7.3 Register an endowment (intake) — Sprint 12

Route: `/en/onboarding` · **Who:** `matrix-admin@example.test` (needs *both* `endowment:waqf:write` and `admin:access_matrix:write` on a **sibling** endowment of the same client — that is what "the family QMULATE already acts for" means in code).

QMULATE PORTFOLIO Select an endowment

الغربية English Sign out

Register an endowment

Records a new endowment for a family QMULATE already acts for: its founder, deed identity, the founder's conditions as stated, the trusteeship appointment, and its first Nazir. The three onboarding gates open with it.

This environment holds fixture data only: identifiers must start with FAKE- and e-mails end with @example.test.

CLIENT (FAMILY)
Al-Rashidi Family (fictional)

Founder (waqif)
☒ An existing founder
Ibrahim Al-Rashidi (fictional)

☐ A new founder
FOUNDER NAME (ARABIC)
FOUNDER NAME (ENGLISH, OPTIONAL)

CERTIFICATE NUMBER DEED NUMBER

WAQF TYPE NATURE
Family (dhurri) In kind (ayni)

ORDER OF ENTITLEMENT FISCAL YEAR END (MM-DD)
Lineage continuation (default) 12-31

CERTIFICATE REGISTRATION DATE
dd/mm/yyyy

FOUNDER'S CONDITIONS, AS STATED IN THE DEED (ARABIC)

Captured once and never edited. What the deed does not say is recorded as unspecified; the distribution engine halts until the deed terms are recorded.

Trusteeship appointment
PRIMARY NAZIR (AS NAMED IN THE DEED) APPOINTMENT DATE
dd/mm/yyyy

☐ Jointly and severally liable representative named
Eligibility criteria as stated (unverified — verify vs primary law)
☒ Muslim

Matrix admin — intake form

Steps:

1. Sign in as `matrix-admin@example.test`, open **Register an endowment** in the sidebar. If the seat had no intake authority, the page would show a notice and **no form** — try it as `nazir@` to see the negative.
2. Client: *Al-Rashidi Family (fictional)*. Founder: pick an existing founder, or *A new founder* and type an Arabic name.
3. Certificate and deed numbers **must start with FAKE-** in this environment (e.g. `FAKE-CERT-MANUAL-1`, `FAKE-DEED-MANUAL-1`). Anything else is refused with `FIXTURE_ONLY_IDENTIFIER_REFUSED` — that refusal is the residency guardrail reaching the UI.
4. Type *Family (dhurri)*, nature *In kind*, order of entitlement, fiscal year end, registration date.
5. Founder's conditions as stated in the deed (Arabic). This is written **once**; what the deed does not say is recorded as *unspecified*, and the engine will halt on this endowment until the deed terms are recorded.
6. Trusteeship: primary Nazir as named in the deed, appointment date, eligibility attestations (unverified — verify vs primary law).
7. First Nazir seat: a staff e-mail ending in `@example.test`, e.g. `nazir@example.test`. **You cannot seat yourself** — entering your own e-mail is refused with `SEGREGATION_OF_DUTIES`.

8. **Register the endowment.** You return to this page with a notice carrying the new endowment's id.

Expected: the endowment is born `NOT_CLASSIFIED` , with **three OPEN onboarding gates** and both gated activities (distribution runs, Authority filings) blocked. The named Nazir now holds a seat on it; **you do not** — the matrix admin was not seated on the newborn, so `/en/endowments/<newId>` answers not-found for you (non-disclosure by design), while `nazir@` can open it and sees its gates at `/en/endowments/<newId>/onboarding` .

Arabic form: `/ar/onboarding` .

The screenshot shows the Arabic onboarding form for a new endowment in the QMULATE system. The form is titled "تسجيل وقفية" (Register Endowment) and includes the following sections:

- الملاحظة** (Note): A section for additional information or comments.
- العمل (الأسرة)** (Family Information): Fields for "عائلة الراشدي (بيانات وهمية)" (Adoptive Family (Fictitious Data)) and "الوقف" (Endowment).
- الوقف** (Endowment): Fields for "وقف قائم" (Existing Endowment), "وقف جديد" (New Endowment), "اسم الوقف (بالعربية)" (Endowment Name (Arabic)), and "اسم الوقف (بالإنجليزية، اختياري)" (Endowment Name (English, Optional)).
- رقم الشهادة** (Certificate Number): A field for the certificate number.
- نوع الوقف** (Type of Endowment): A dropdown menu with options like "عيني" (Physical) and "أهلي (ذري)" (Personal/Hereditary).
- ترتيب الاستحقاق** (Priority of Claim): A dropdown menu with options like "استمرار النسب (الأصل)" (Continuation of Lineage (Original)).
- تاريخ تسجيل الشهادة** (Date of Certificate Registration): A date field in dd/mm/yyyy format.
- شروط الوقف كما ورد في الصك (بالعربية)** (Conditions of the Endowment as stated in the Deed (Arabic)): A large text area for the deed conditions.
- صك النظارة** (Deed of Endowment): Fields for "تاريخ التعيين" (Date of Appointment) and "النظر الأصلي (كما سُقي في الصك)" (Original Deed (as stated in the Deed)).
- الشروط الإلزامية** (Mandatory Conditions): A section with checkboxes for "سُقي ويكيل معروض مسؤول بالتأمين والتكاليف" (Insured and financed by the responsible party), "شروط الأهلية كما وردت (غير متحقق منها - راجع النظام الأصلي)" (Eligibility conditions as stated (not met - see original system)), "مسلم" (Muslim), "كامل الأهلية" (Full eligibility), and "لم يسبق عزله بما يُسقط أهليته" (Not previously removed by what nullifies eligibility).

Matrix admin — intake, Arabic

Automated: `onboarding-intake.spec.ts` ; `onboarding-intake.integration.test.ts` (api); `endowment-birth-admission.integration.test.ts` (database: the `waqf_birth_admission` trigger, born-retired refusal, sibling authority).

7.4 Onboarding gates — the three-gate state machine (BR-1101, V-11)

Route: `/en/endowments/waqf-004/onboarding` · **Who:** `clerk@example.test` clears; `nazir@example.test` reopens.

QMULATE PORTFOLIO Select an endowment العربية English Sign out

ENDOWMENT Small
FAKE-1000004
 Founder (waqf): Aisha Al-Rashidi (fictional)

RECORD TRUSTEESHIP DEED CLASSIFICATION FOUNDER'S CONDITIONS BENEFICIARIES RESERVED MATT

Onboarding gates
 Three sequenced handover gates. Nothing downstream proceeds until the prior gate clears: a distribution run and an Authority filing are blocked until Gate 02 (Systems & Controls) is cleared.

Blocked now
 • distribution runs • Authority filing submissions

Gate 01 — Authority & Legal Cleared
 Cleared by user-seed-admin
 Cleared on Feb 1, 2026 (1447-08-13)

Reopen this gate
 Only the Nazir may reopen a cleared gate. Reopening blocks downstream activity again; the reason is recorded.
 REASON *

 Reopen the gate

Gate 02 — Systems & Controls Open
 While open, this gate blocks
 • distribution runs • Authority filing submissions

Clear this gate
 Attest each item as done. The gate is cleared over the facts on record; an attestation cannot stand in for a missing deed, classification, account or registry.

The clearer attests
☐ IDENTITY AND EMAIL STOOD UP
☐ CLOUD ACCOUNTING ON THE SOCPA CHART
☐ DEDICATED BANK ACCOUNTS OPENED
☐ OPERATIONS WORKSPACE AND COMPLIANCE CALENDAR
☐ VAULT AND DASHBOARD STOOD UP
 NOTE (OPTIONAL)

 Clear the gate

Nazir — waqf-004 onboarding gates

The seed leaves `waqf-004` with **Gate 01 cleared, Gate 02 open, Gate 03 open**. The page states what is *blocked now* (distribution runs, Authority filing submissions) and, per gate, who cleared it and when.

Clear Gate 02 as the clerk:

1. Sign in as `clerk@example.test`, open the route above.
2. On *Gate 02 — Systems & Controls*, tick all **five** attestations (identity and e-mail, cloud accounting on the SOCPA chart, dedicated bank accounts, operations workspace and compliance calendar, vault and dashboard). The *Clear the gate* button enables only when all five are ticked.
3. Save. The notice reads *gate cleared*; Gate 02 shows **Cleared by user-reserved-clerk-001**; *Blocked now* becomes *nothing is blocked*; Gate 03's form now appears (it was refused with *the prior gate has not been cleared before*).

Order is enforced both ways: you cannot clear Gate 03 before Gate 02, and reopening Gate 02 re-blocks downstream activity.

Reopen as the Nazir: as `nazir@`, a cleared gate shows *Reopen this gate* with a mandatory **reason**. The clerk sees no reopen form — *only the Nazir may reopen a cleared gate* (owner ruling: "nazir can reopen").

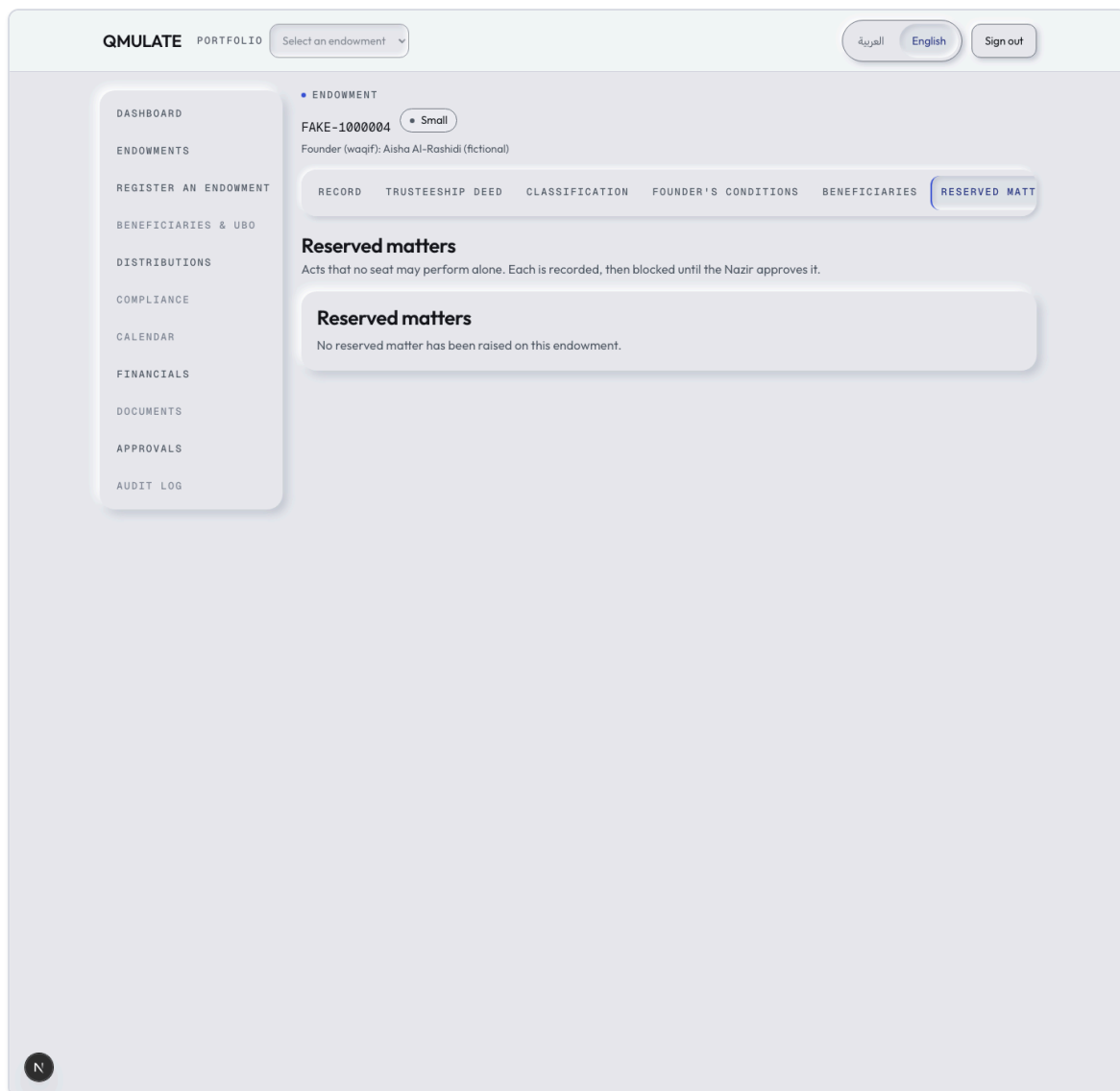
V-11 by hand: with Gate 02 open, an Authority filing submission or a distribution-run submission on `waqf-004` is refused with `ONBOARDING_GATE_NOT_CLEARED` ("This endowment has not cleared onboarding Gate 02 (Systems & Controls), so distributions and Authority filings..."). Clear the gate, repeat, and it goes through.

Automated: `onboarding-gate.spec.ts` (V-11 in the browser), `packages/api/test/onboarding-gate.integration.test.ts`, `packages/database/test/onboarding-gate.integration.test.ts` (order both directions, no delete/truncate, trigger census).

7.5 Reserved matters — the BR-1102 chain

Route: `/en/endowments/waqf-004/reserved-matters` · **Who:** `clerk@example.test` records the chain steps; the **sign is the Nazir's**.

The nine reserved-matter kinds are `ASSET_DISPOSAL`, `ASSET_SUBSTITUTION_ISTIBDAL`, `ASSET_PLEDGE`, `ASSET_LONG_LEASE`, `DEED_IDENTITY`, `DEED_TERM_RECORD`, `ACCESS_MATRIX_CHANGE`, `CLASSIFICATION_RETURN_TO_NOT_CLASSIFIED`, `RECEIPT_CLASS_CORRECTION`. The Shart al-Waqif is **not** among them: there is no kind that opens the founder's conditions, by design.



Clerk — reserved matters, none raised

Nothing is raised in the fixture. **There is no UI to raise a reserved matter yet** (§9); the E2E journey raises one through the API and so must you. While signed in as the clerk, paste this into the browser console on any app page:

```
await fetch('/api/trpc/approval.initiate', {
  method: 'POST',
  headers: { 'content-type': 'application/json', 'x-qmulate-locale': 'en' },
  body: JSON.stringify({
    waqfId: 'waqf-004',
    type: 'RESERVED_MATTER',
    reservedMatterKind: 'DEED_IDENTITY',
    subjectId: 'waqf:waqf-004>manual:' + Date.now(),
    payload: { kind: 'manual.test', waqfId: 'waqf-004', locale: 'en', note: 'manual test
    (بيانات وهمية)' },
  })),
}).then((r) => r.json());
```

Expected on the page after a reload:

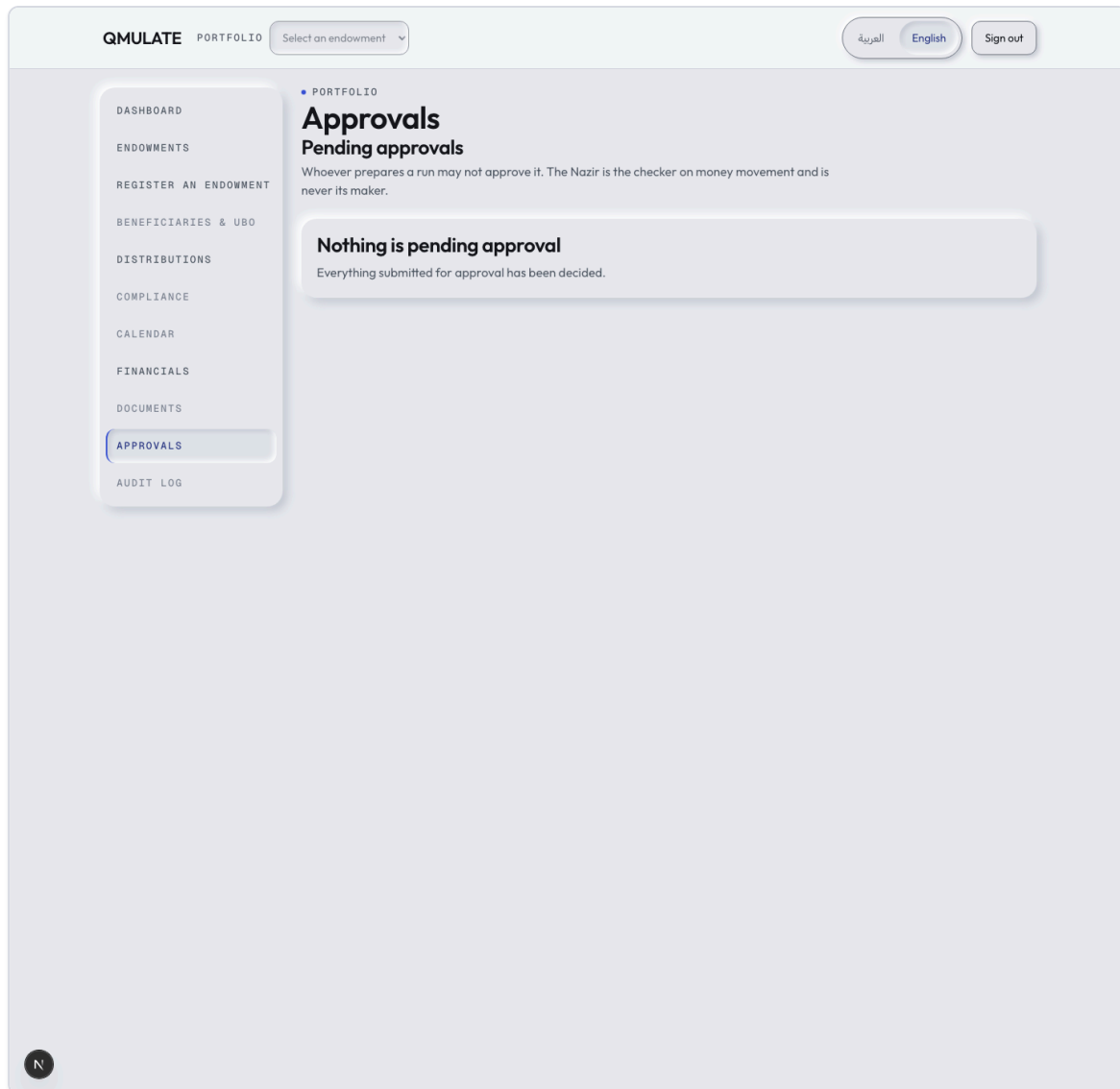
1. A card for the matter, badged **Blocked**, stating *chain enforced*, and a *sign blocked* panel naming the two missing steps: **Principal consent** and **Counsel review**.
2. One form per missing step. Enter a reference (a document or minute number, `FAKE-...`) and save. The notice reads *chain step recorded*; that step shows as recorded with who and when.
3. After both steps, the panel flips to *sign ready*. **The clerk still cannot sign** — the sign is the Nazir's approval, and the person who raised the matter cannot approve it (`SEGREGATION_OF_DUTIES`). Sign in as `nazir@` to see the ready state; approval happens through the approvals path (§7.7's approve verb).

Database wall behind the screen: even a privileged direct write cannot execute a reserved matter without all three steps recorded (migration 51). The chain is a wall, not a form validation.

Automated: `reserved-chain.spec.ts`; `reserved-matter-chain.integration.test.ts` (database); the `reserved-matter-surface` and `labels` pins (9 of 9 kinds in both catalogues).

7.6 Approvals queue

Route: `/en/approvals` · **Who:** any seat with `approval:request:read` (nazir, both admins, board, auditor, finance...).



Nazir — approvals queue

Empty **by fact** in the fixture: the seeded `dist-001` approval is already terminal. The queue fills when the accountant submits a run (§7.7) or a reserved matter is raised (§7.5). Each row names the maker; a row you raised yourself is marked and cannot be approved by you.

7.7 Distribution run — compute, submit, approve, execute

Routes: `/en/distributions` → `/en/distributions/waqf-001` → `/new?periodStart=2026-01-01&periodEnd=2026-03-31&step=period|waterfall|lines|review` · **Who:** `accountant@example.test` computes and submits; `nazir@` or `approver@` approves and executes.

The wizard's whole state is in the URL, so a step is a navigation. Sign in as the accountant and walk `waqf-001` for the period 2026-01-01 → 2026-03-31:

1. **Period** — the fiscal window and the 3-month post-FYE deadline (unverified figure).

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English

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Compute a distribution run

waqf-001

Only ghallah (income) is distributed. The endowment's corpus enters no figure on these screens and is never added to income in one total.

PERIOD

WATERFALL

LINES

REVIEW

Period

The period's receipts are read from the ledger with the classification recorded at entry; nothing is classified here.

This is a preview and is not written to the record: no lines and no entries until the run is saved.

Period

The period's receipts are read from the ledger with the classification recorded at entry; nothing is classified here.

PERIOD START *

01/01/2026

PERIOD END *

31/03/2026

Recompute

This is a preview and is not written to the record: no lines and no entries until the run is saved.

CORPUS

Corpus receipts — held out of the distribution

The corpus, and whatever stands in its place, is never distributed. Sale proceeds and istibdal proceeds are corpus, not income: they return to the corpus and enter none of the waterfall figures above. They are shown here as visibly excluded rather than absent.

CORPUS RECEIPTS

1

CORPUS RECEIPTS TOTAL

SAR 4,200,000.00

This amount is never added to income revenue in a combined total.

rev-005 SAR-4,200,000.00

Corpus istibdal (substitution) proceeds

NEXT

Accountant — period

2. **Waterfall** — *ṣiḡāna* reserved **first**, then operating costs, then the Nazir fee (10 % of revenue by *this* deed, not statute), then the distributable. Only income enters. Check the header sentence: "*Only ghallah (income) is distributed. The endowment's corpus enters no figure on these screens.*"

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ENDOWMENT

Compute a distribution run

waqf-001

Only ghalah (income) is distributed. The endowment's corpus enters no figure on these screens and is never added to income in one total.

PERIOD

WATERFALL

LINES

REVIEW

Waterfall

The order is binding: maintenance first, then operating costs, then the Nazir fee, then the distributable.

This is a preview and is not written to the record: no lines and no entries until the run is saved.

CORPUS

Corpus receipts — held out of the distribution

The corpus, and whatever stands in its place, is never distributed. Sale proceeds and istibdal proceeds are corpus, not income: they return to the corpus and enter none of the waterfall figures above. They are shown here as visibly excluded rather than absent.

CORPUS RECEIPTS

1

CORPUS RECEIPTS TOTAL

SAR 4,200,000.00

This amount is never added to income revenue in a combined total.

zev-005 SAR-4,200,000.00

Corpus - Istibdal (substitution) proceeds

INCOME

The distribution waterfall

The order is binding and nothing may be re-ordered: maintenance is reserved from income before any cost, then operating costs, then the Nazir fee; what remains is the distributable.

INCOME REVENUE

SAR 350,000.00

MAINTENANCE RESERVED

SAR 40,000.00

OPERATING COSTS

SAR 0.00

NET INCOME

SAR 310,000.00

NAZIR FEE

SAR 35,000.00

Percentage of revenue

The Nazir fee is set by the waqf deed — not by statute and not by a system setting.

Unverified figure — confirm against primary law

DISTRIBUTABLE

SAR 275,000.00

Accountant — waterfall

- Lines** — one line per beneficiary with **how this line was decided**: rule, tier, reason. On waqf-001 :
 - ben-001 *Paid*, ben-002 *Excluded* (UPPER_TABAQA_EXTANT — a lower tier while an upper one lives), ben-003 **Withheld** (KYC_UNVERIFIED ; G-4 — the line exists, the payment does not).

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Compute a distribution run

waqf-001

Only ghallah (income) is distributed. The endowment's corpus enters no figure on these screens and is never added to income in one total.

PERIOD

WATERFALL

LINES

REVIEW

Lines

One line per recorded beneficiary, each with the reason it was withheld or excluded, if any.

This is a preview and is not written to the record: no lines and no entries until the run is saved.

DISTRIBUTION LINES

Distribution lines

A procedural hold stops payment without reducing entitlement: the amount stands unchanged.

BENEFICIARY

ben-001

PAID

ENTITLEMENT

SAR 137,500.00

SHARE

50.000000%

HOW THIS LINE WAS DECIDED

Entitlement rule

The beneficiary is entitled as a member of the highest generation that still has living entitled members, according to the order set out in the deed. Within that generation, each share follows the weight recorded in the deed and is not automatically equal.

ORDERED_LOWEST_LIVING_TABAQA

Endowment

Family member

Tier

1

Reason

Not applicable

BENEFICIARY

ben-002

EXCLUDED

ENTITLEMENT

SAR 0.00

SHARE

0.000000%

HOW THIS LINE WAS DECIDED

Entitlement rule

The beneficiary is entitled as a member of the highest generation that still has living entitled members, according to the order set out in the deed. Within that generation, each share follows the weight recorded in the deed and is not automatically equal.

ORDERED_LOWEST_LIVING_TABAQA

Endowment

Tier

2

Reason

Accountant — lines

- Review** — the corpus receipt **rev-005 (istibdal proceeds, SAR 4,200,000)** visibly held out of the distribution (V-1); totals; the engine version and run digest; the entitlement rule; the invariants asserted and — separately — the invariants this run makes **no claim about**; a computation trace step by step. *"This is a preview and is not written to the record."*

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QMULATE PORTFOLIO Select an endowment المربية English Sign out

ENDOWMENT

Compute a distribution run waqf-001

Only ghallah (income) is distributed. The endowment's corpus enters no figure on these screens and is never added to income in one total.

PERIOD WATERFALL LINES REVIEW

Review

The figures and notes are reviewed before submission. Computing alone pays nothing.
This is a preview and is not written to the record: no lines and no entries until the run is saved.

CORPUS

Corpus receipts — held out of the distribution

The corpus, and whatever stands in its place, is never distributed. Sale proceeds and istibdal proceeds are corpus, not income: they return to the corpus and enter none of the waterfall figures above. They are shown here as visibly excluded rather than absent.

CORPUS RECEIPTS	CORPUS RECEIPTS TOTAL
1	SAR 4,200,000.00

This amount is never added to income revenue in a combined total.

rev-005 SAR-4,200,000.00 Corpus - Istibdal (substitution) proceeds

INCOME

The distribution waterfall

The order is binding and nothing may be re-ordered: maintenance is reserved from income before any cost, then operating costs, then the Nazir fee; what remains is the distributable.

INCOME REVENUE	
	SAR 350,000.00
MAINTENANCE RESERVED	SAR 40,000.00
OPERATING COSTS	SAR 0.00
NET INCOME	SAR 310,000.00
NAZIR FEE	SAR 35,000.00
	Percentage of revenue
	The Nazir fee is set by the waqf deed — not by statute and not by a system setting.
	⚠ Unverified figure — confirm against primary law
DISTRIBUTABLE	SAR 275,000.00

Accountant — review

- Submit** the run. Its status becomes *pending approval*; it appears in the approvals queue and in the KPI tile *Runs awaiting approval*. **You cannot approve it** as the accountant (no `distribution:run:approve`).
- Sign in as `approver@example.test` (or `nazir@`), open the run from `/en/distributions/waqf-001` and **approve**. Try approving as the same person who submitted: `SEGREGATION_OF_DUTIES` — "You raised this request, so you cannot approve it. It needs a different Nazir." (G-3).
- Execute** as a Nazir seat. The run becomes *executed*; the maker/checker panel names two different identities.

The seeded historical run shows what a stored run looks like: `/en/distributions/waqf-001/runs/dist-001` — maker `user-accountant-001`, checker `user-approver-001`, the stored waterfall and lines, and a trace that records the fixture's own internal inconsistency rather than repairing it.

QMULATE

PORTFOLIO

Select an endowment

العربية

English

Sign out

DASHBOARD

ENDOWMENTS

REGISTER AN ENDOWMENT

BENEFICIARIES & UBO

DISTRIBUTIONS

COMPLIANCE

CALENDAR

FINANCIALS

DOCUMENTS

APPROVALS

AUDIT LOG

DISTRIBUTIONS

Distribution run

PERIOD Jan 1, 2026 (1447-07-12) - Mar 31, 2026 (1447-10-12)

MAKER AND CHECKER

Maker and checker

Whoever prepares a run may not approve it. The Nazir is the checker on money movement and is never its maker.

Executed

Executed

PREPARED BY

user-accountant-001

APPROVED BY

user-approver-001

RUN DIGEST

7b9ee80b29fa7db0bb8d236fb45c6499fc5127ab995ee538c4b7b838902184cc

A fingerprint computed over the run's inputs and result — this is what the Nazir approves. A different engine version makes two runs incomparable.

Payment lines are written only against an approval that names this very run, with a maker who is not the checker.

The distribution waterfall

The order is binding and nothing may be re-ordered: maintenance is reserved from income before any cost, then operating costs, then the Nazir fee; what remains is the distributable.

INCOME REVENUE	SAR 350,000.00
MAINTENANCE RESERVED	SAR 40,000.00
OPERATING COSTS	SAR 0.00
NAZIR FEE	SAR 35,000.00
The Nazir fee is set by the waqf deed — not by statute and not by a system setting. ⚠ Unverified figure — confirm against primary law	
DISTRIBUTABLE	SAR 275,000.00

CALCULATION ENGINE VERSION

Not recorded

RUN DIGEST

Not recorded

SUBMITTED

Apr 20, 2026 (1447-11-03)

A fingerprint computed over the run's inputs and result — this is what the Nazir approves. A different engine version makes two runs incomparable.

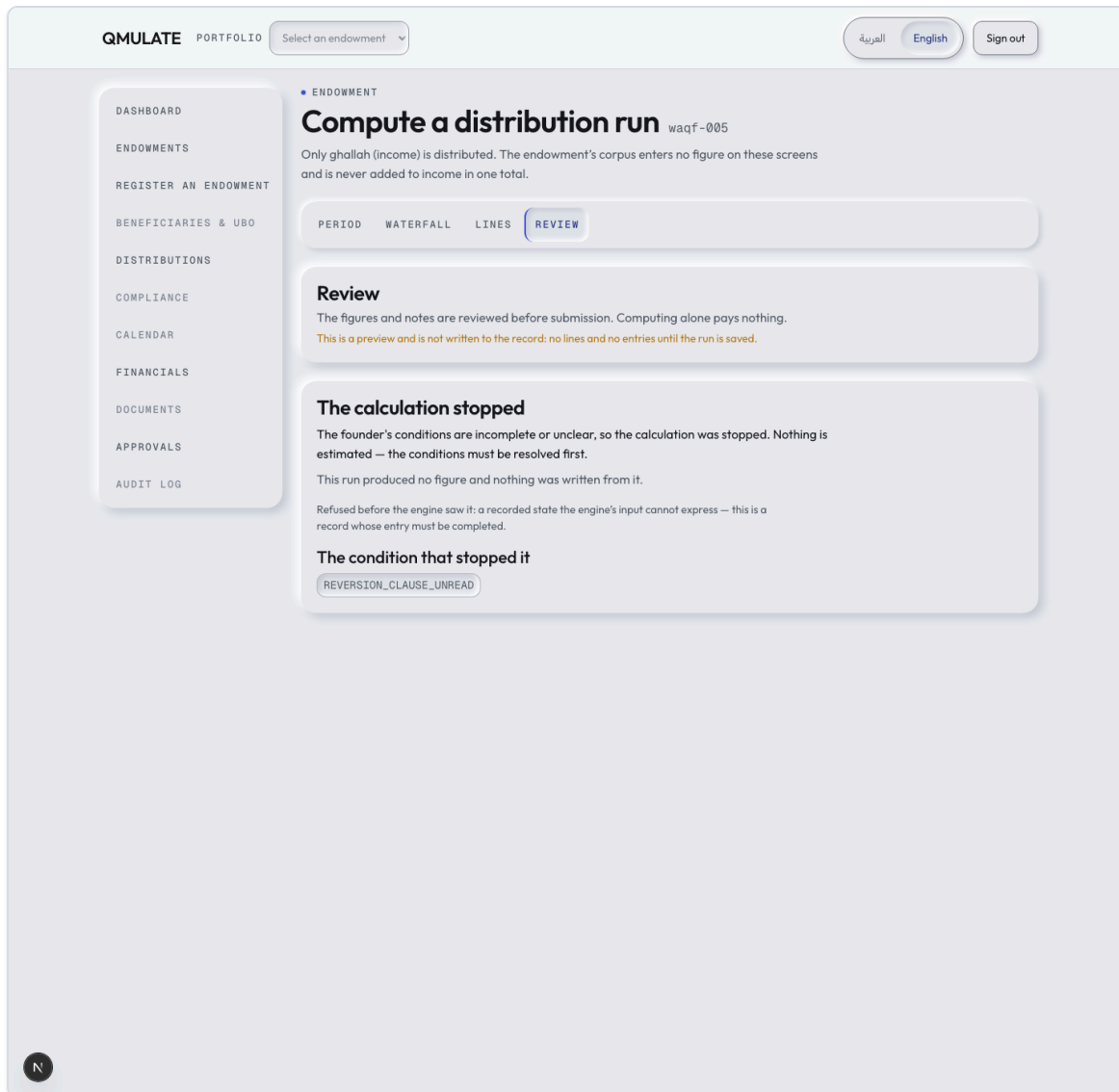
Nazir — stored run dist-001

Two refusals worth seeing:

- waqf-005 halts: /en/distributions/waqf-005/new?...&step=review → The calculation stopped — REVERSION_CLAUSE_UNREAD , "Nothing is estimated — the conditions must be resolved first." SHART_INCOMPLETE in the product's words: the engine never guesses (Binding rule 1).

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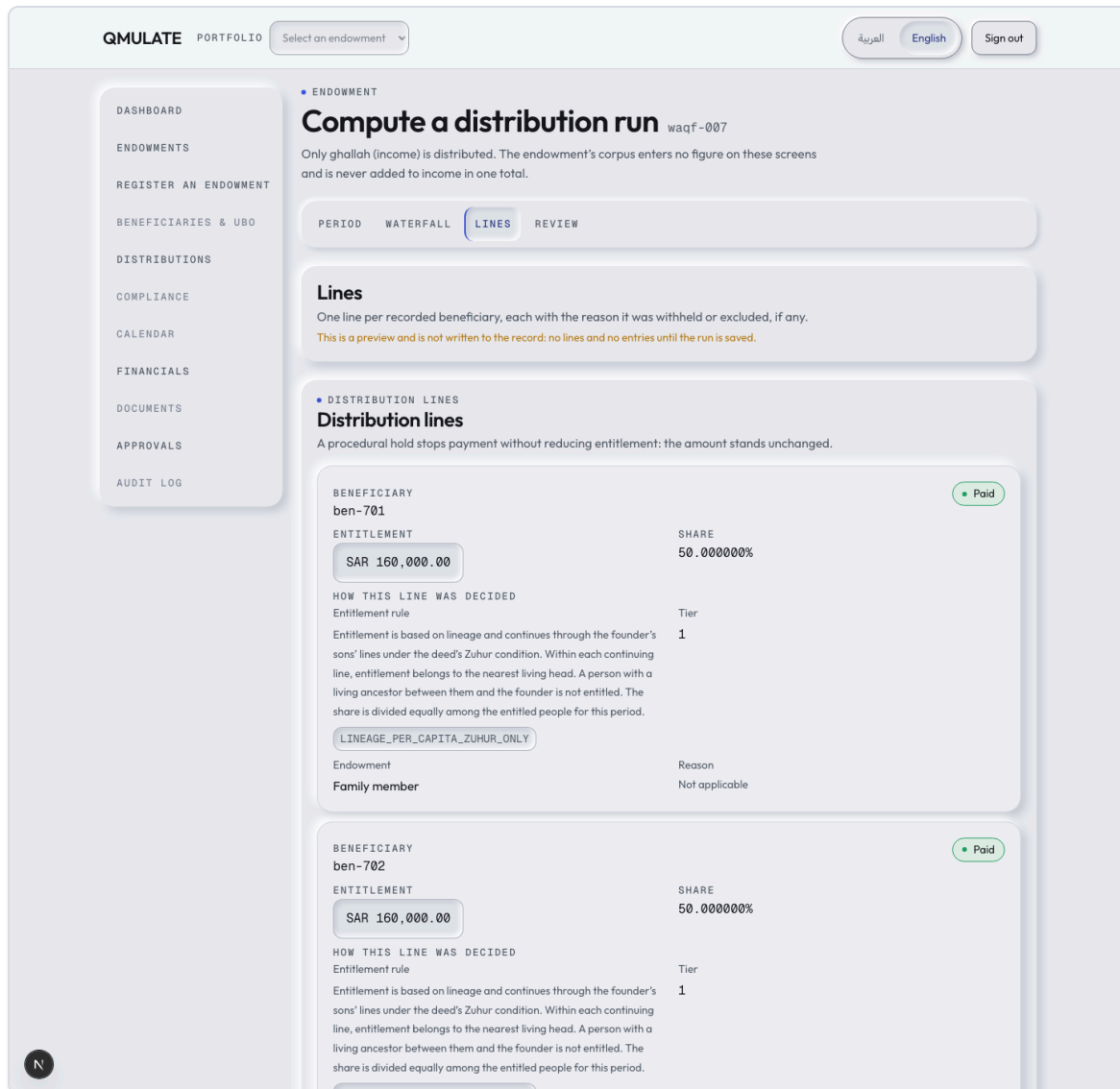
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Accountant — waqf-005 halts

- waqf-004 while Gate 02 is open: submitting is refused with ONBOARDING_GATE_NOT_CLEARED (V-11).

Lineage deed: waqf-007 's lines step shows the default lineage rule — per capita over the living frontier of each line, the approved wording.



Accountant — waqf-007 lineage lines

Negative control: as `unscoped@` every distribution surface is not-found; signed out, every one redirects to sign-in.

Automated: `distribution.spec.ts` (6 journeys), the ~2,160-test domain suite, `distribution-run-schema` and `approval-decision-plane` integration suites.

7.8 Financials

Route: `/en/financials`, `/en/financials?waqf=<id>` · **Who:** `auditor@example.test` (read-only: two verbs of seventeen), `nazir@`, `accountant@`.

Every blended figure sits beside its **corpus companion**; capital receipts are counted and totalled but never blended into income; a direct-use endowment (`waqf-004`) *states* direct utilization instead of rendering zeros; anything not modelled in this release is declared *owed* (reconciliation, statements) rather than shown as a zero.

QMULATE

PORTFOLIO

Select an endowment

العربيةEnglishSign out

DASHBOARD

ENDOWMENTS

REGISTER AN ENDOWMENT

BENEFICIARIES & UBO

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COMPLIANCE

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PORTFOLIO

Financials

Cash, receipts, expenses and distributions for one endowment — read only.

Endowment

One endowment at a time: every figure below is scoped to the one selected.

FAKE-1000001FAKE-1000003FAKE-1000004FAKE-1000005FAKE-1000007

Cash position

FAKE-1000001 SAR 4,475,000.00 of which corpus (asl) — never distributable SAR 4,200,000.00

FAKE-1000003 SAR 21,680,000.00 of which corpus (asl) — never distributable SAR 20,000,000.00

FAKE-1000004 SAR 0.00 of which corpus (asl) — never distributable SAR 0.00

FAKE-1000005 SAR 0.00 of which corpus (asl) — never distributable SAR 0.00

FAKE-1000007 SAR 400,000.00 of which corpus (asl) — never distributable SAR 0.00

CASH HELD

SAR 4,475,000.00

of which corpus (asl) — never distributable SAR 4,200,000.00

A net balance mixes income and corpus by construction; the corpus figure beside it says how much may never be distributed.

INCOME (GHALLAH) — THE ONLY CLASS THAT MAY ENTER THE WATERFALL

SAR 350,000.00

Capital (corpus) — blocked from every distribution

SAR 4,200,000.00

Income and capital are separate classes and are never summed into a distributable figure.

Dedicated accounts

FAKE-ACCT-W1 Dedicated

Net on this account SAR 4,475,000.00 of which corpus SAR 4,200,000.00

A net balance mixes income and corpus by construction; the corpus figure beside it says how much may never be distributed.

Capital by source

ISTIBDAL_PROCEEDS

SAR 4,200,000.00

Expenses

Total expenses SAR 75,000.00

MAINTENANCE SAR 40,000.00

NAZIR_FEE SAR 35,000.00

Distributions

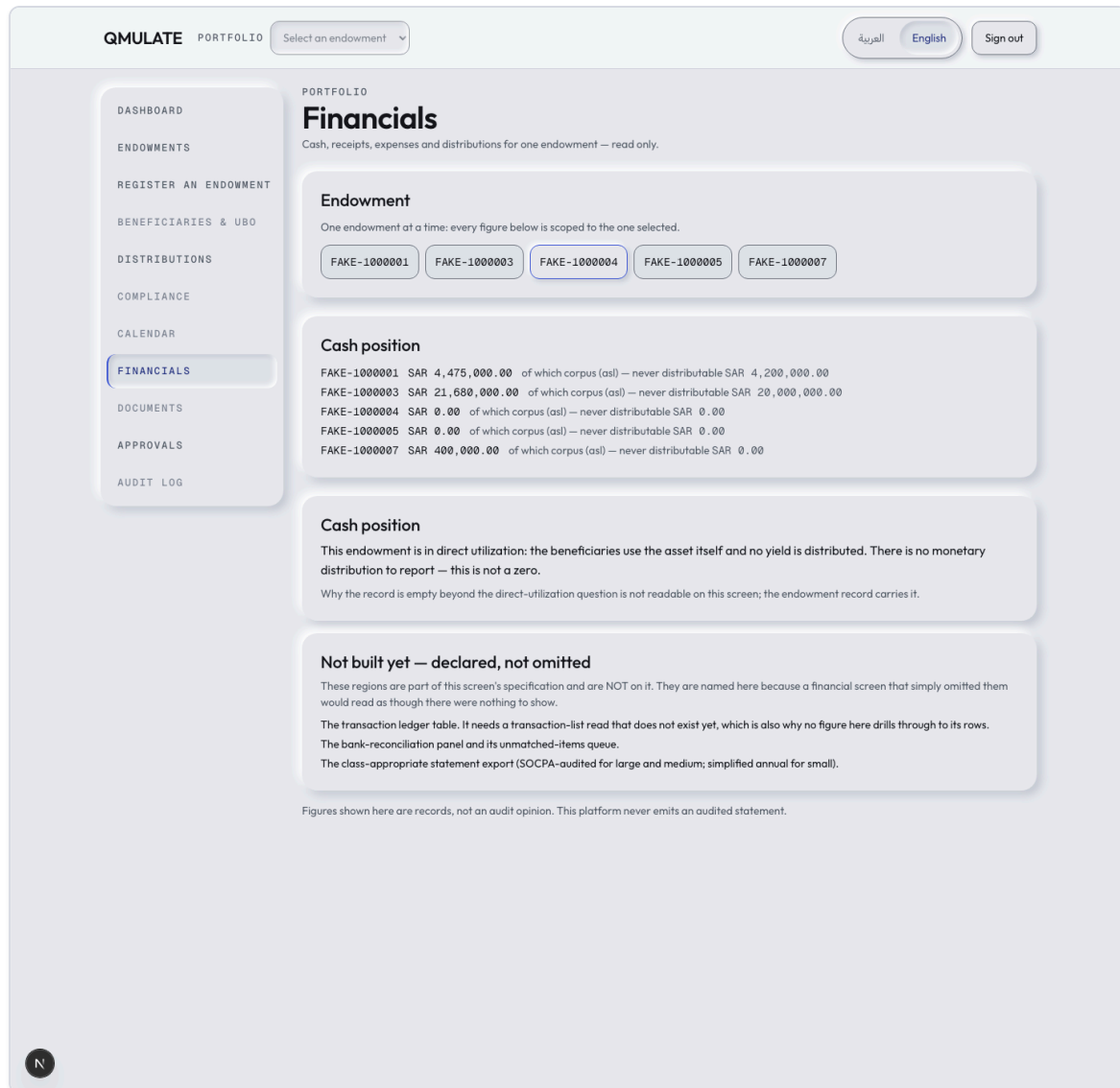
1 recorded

Executed distributable

SAR 275,000.00

Arrears

Not tracked yet — this is not a zero. Nothing in the record states what is owed and unpaid, so no arrears figure can be shown.



Auditor — financials waqf-004, direct use

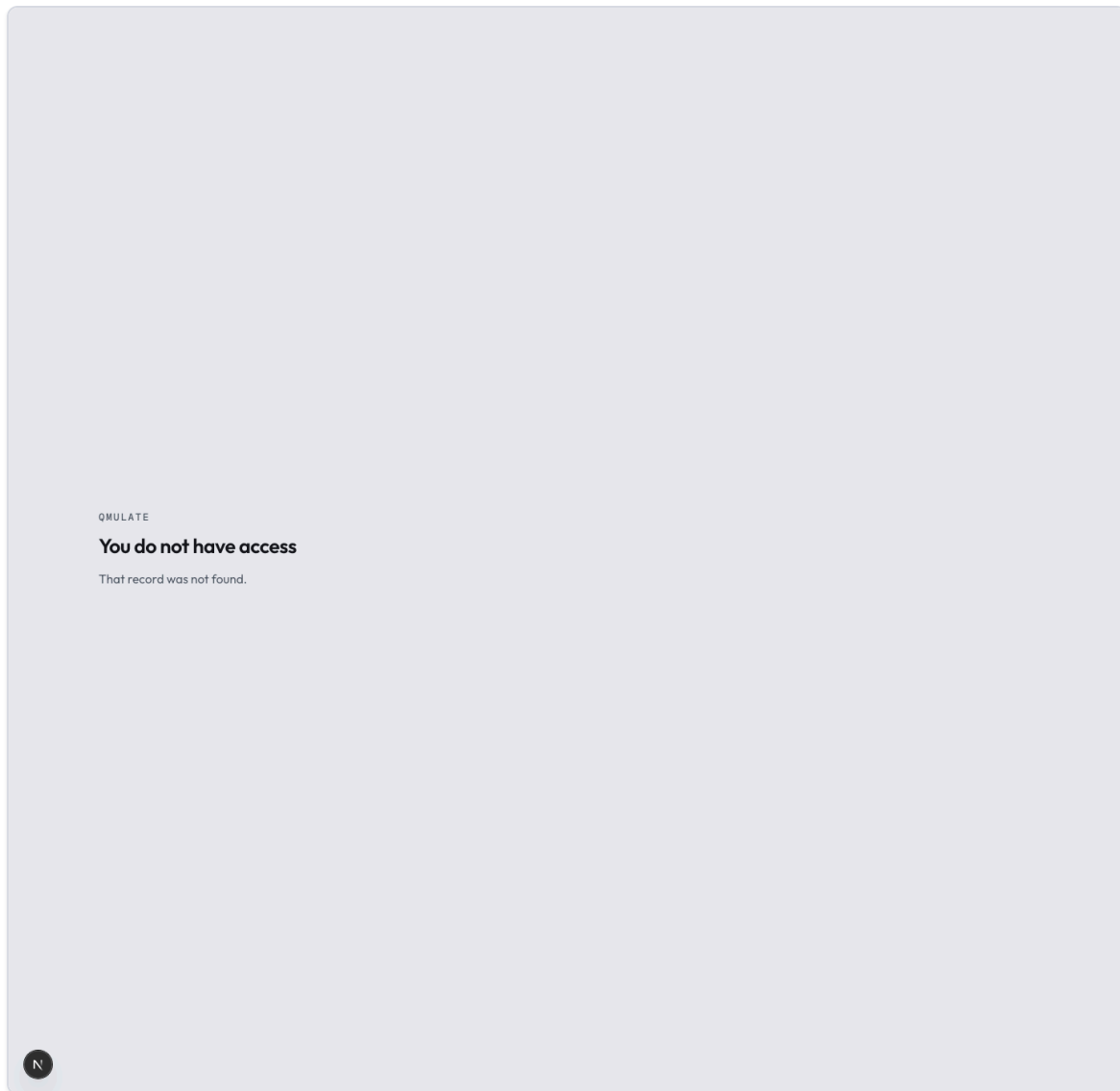
Receipt capture is API-only (there is no finance capture form yet, §9). Every **REVENUE** row must carry a class — income or capital — at entry; the database refuses an unclassified receipt.

Automated: `financials-journey.spec.ts` ; the ledger's integration suites (G-2 composite FKs, receipt-class CHECK).

7.9 Access matrix — deny by default (G-7)

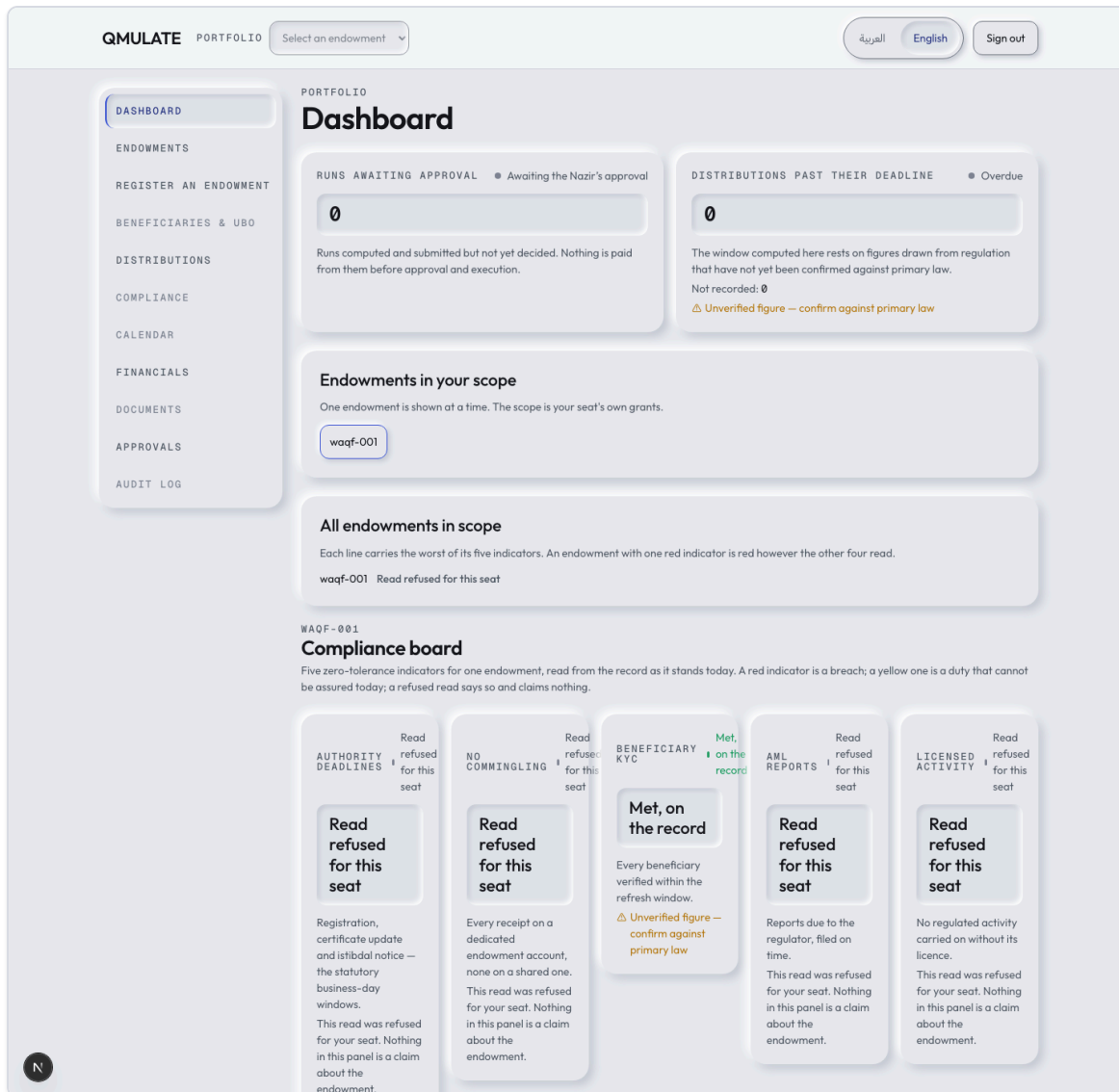
Run these three in sequence; they are the cheapest security check in the product.

1. **unscoped@example.test** — a valid, TOTP-enrolled session with zero grants. `/en/dashboard` and `/en/endowments` both answer **"You do not have access — That record was not found."** Nothing about the portfolio is disclosed, not even that endowments exist.



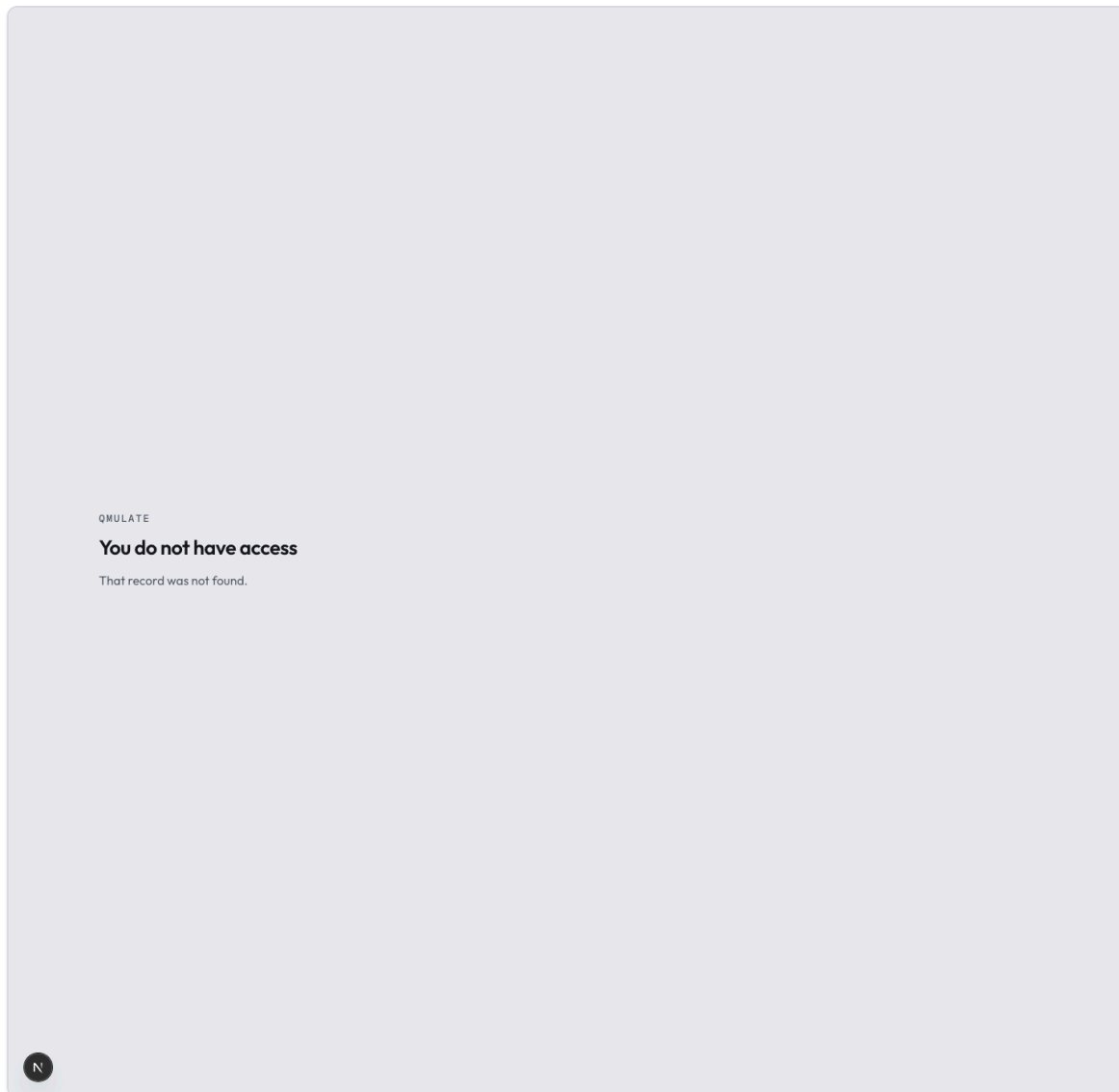
Unscoped — dashboard

2. **beneficiary.ben-001@example.test** — seated on **waqf-001** for its own record. The dashboard shows exactly one endowment in scope and **"Read refused for this seat"** on four of five indicators; **/en/endowments/waqf-001/deed** reads (owner ruling Q-E4-1); every other endowment is not-found; the ledger and co-beneficiaries are never reachable.



Beneficiary — dashboard

3. **admin@example.test** — platform administrator with **no endowment grant**. Same not-found on `/en/endowments` : administrative authority is not a grant, and nothing in the access matrix is satisfied by "the admin has a grant".



Seed admin — endowments

Positive control for all three: `nazir@` sees the whole hierarchy on the same routes (§7.2).

Automated: `kernel.spec.ts`, `endowment.spec.ts`, `endowment-journey.spec.ts`, `grant-escalation`, `router-introspection` (the UNSCOPED mutation allowlist is exactly `onboarding.intake`), `base-client-export-surface`.

7.10 Locale and RTL

The toggle in the top bar switches `ar` ↔ `en` **preserving the path**. Arabic is the product default. Check: `dir="rtl"` on the root, mirrored layout, no horizontal scroll at any width, focus is a solid ring, the skip link is the first tab stop, numerals and IDs stay LTR islands.

Automated: `shell.spec.ts`, `auth.spec.ts`, the i18n parity and Arabic-floor unit suites (every key exists in both catalogues; the Arabic catalogue may not shrink).

8 · The importer and the residency guardrail (G-8, V-12) — CLI only

There is no UI. The importer is the fourth layer of the residency guardrail and refuses before it reads anything:

```
# Layer 1 – env: production is only legal beside DATA_RESIDENCY=ksa
DATA_CLASSIFICATION=production pnpm --filter @qmulate/database run import --source
./some.json
# → refused: DATA_RESIDENCY must be "ksa" when DATA_CLASSIFICATION is "production"

# Layer 2 – seed refuses anything but fixture-only
DATA_CLASSIFICATION=production pnpm --filter @qmulate/database run seed
# → refused

# Layer 3 – importer refuses a non-production / non-KSA target, the fixture itself as a
source,
#           and any source carrying a fictional marker ("Entirely fictional", "بيانات وهمية",
"(fictional)")
DATA_CLASSIFICATION=fixture-only pnpm --filter @qmulate/database run import --source
data/fixtures/sample-waqf.json
# → IMPORT_REFUSED

# Layer 4 – CI proves all of it: steps G-8a...G-8f in the "Residency guardrail (G-8)" job
```

A dry run (`--dry-run`) prints the plan without applying. Applying runs on the owner connection inside one audited transaction as actor `import:<runId>` ; gates are born OPEN; distributions, compliance tasks, settings, users and approvals are deliberately **not** imported.

9 · Coverage gaps and what is manual-only

Honest list, as of this commit. None of these is a hidden defect; each is recorded in

`docs/product/prd/BUILD-PLAN.md` .

Not built (nav items rendered greyed, not linked): a global *Beneficiaries & UBO* screen (per-endowment beneficiaries exist under the record), *Compliance* register screen (the board lives on the dashboard), *Calendar*, *Documents* vault UI (the storage adapter and retention floor exist; the screen does not), *Audit Log* viewer.

API-only today (no form): raising a reserved matter (§7.5), recording receipts and bank movements, issuing access-matrix seats, requesting an Authority filing submission. Each has integration coverage; none has a screen.

TOTP enrolment shows the URI as text, while the copy says "scan the QR code". Works with "enter a setup key"; a QR image is owed.

Owner-owned decisions the tests deliberately do not resolve (Binding rule 4): Q7 option (b) (a brand-new client — the owner-credential importer ships instead), Q8 (a kindless settings change carries no chain), Q4 (the gate checklist wording), the Arabic sign-off on the three S12 screens (intake, gates, reserved matters — their Arabic is drafted, not approved), the E10 P0/P1 exit conflict, the five registration dates, Q3 (maintenance when the deed is silent), zakat, the four edge receipt types.

Known manual-only checks: visual RTL quality beyond what `shell.spec.ts` asserts; the wording of every Arabic screen (product-approved legal text must not be invented in a code change); anything printable (no printable statement exists yet; the HQ city is unresolved and must not be printed).

No performance or load tests exist. No visual-regression suite exists. Accessibility is asserted structurally (hit targets, focus ring, skip link, `lang / dir`), not audited.

10 · Regenerating the screenshots

`docs/testing/screenshots.mts` is the harness that took every image in this folder: it signs in through the real UI as each seat, enrolls TOTP on first use (storing each secret in a scratch JSON so a rerun can pass the challenge), and screenshots a fixed list of routes at 1440 px. With the \$4 stack running on port 3100:

```
SCRATCH=/tmp/qm-shots pnpm --filter web exec tsx ../../docs/testing/screenshots.mts
# every seat
SCRATCH=/tmp/qm-shots pnpm --filter web exec tsx ../../docs/testing/screenshots.mts nazir
# one seat
```

It imports Playwright from `apps/web`'s own dependency tree, so it needs no install of its own. Because TOTP enrolment is one-way, run it against a **fresh** `--reset` **cluster** or keep the scratch secrets file from the run that enrolled the seats.

The printed hand-out (`docs/testing/TESTING-GUIDE.pdf`) is built from this file by `docs/testing/build-pdf.mts`. It embeds *viewport-only* screenshots (1440 × 1400, taken with `FULLPAGE=0 VIEWPORT_H=1400 OUT=<dir>`), because the full-page images in this folder are too tall to print legibly:

```
mkdir -p /tmp/qm-print && cd apps/web
FULLPAGE=0 VIEWPORT_H=1400 OUT=/tmp/qm-print SCRATCH=/tmp/qm-shots pnpm exec tsx
../../docs/testing/screenshots.mts
SHOTS=/tmp/qm-print pnpm exec tsx ../../docs/testing/build-pdf.mts
```

Rebuild the PDF whenever this file or the screens change; the PDF is the copy handed to users, this file is the source.

Sources of truth this document summarises: `docs/product/prd/17-build-ship-dod.md` (gates, scenarios), `docs/product/prd/BUILD-PLAN.md` (the sprint rows and their measurements), `apps/web/e2e/*.spec.ts` (the journeys), `packages/database/src/seed/map.ts` (the seats), `packages/domain/src/access.ts` (the role presets), `.github/workflows/ci.yml` (the jobs). When they disagree with this page, they win — update this page.